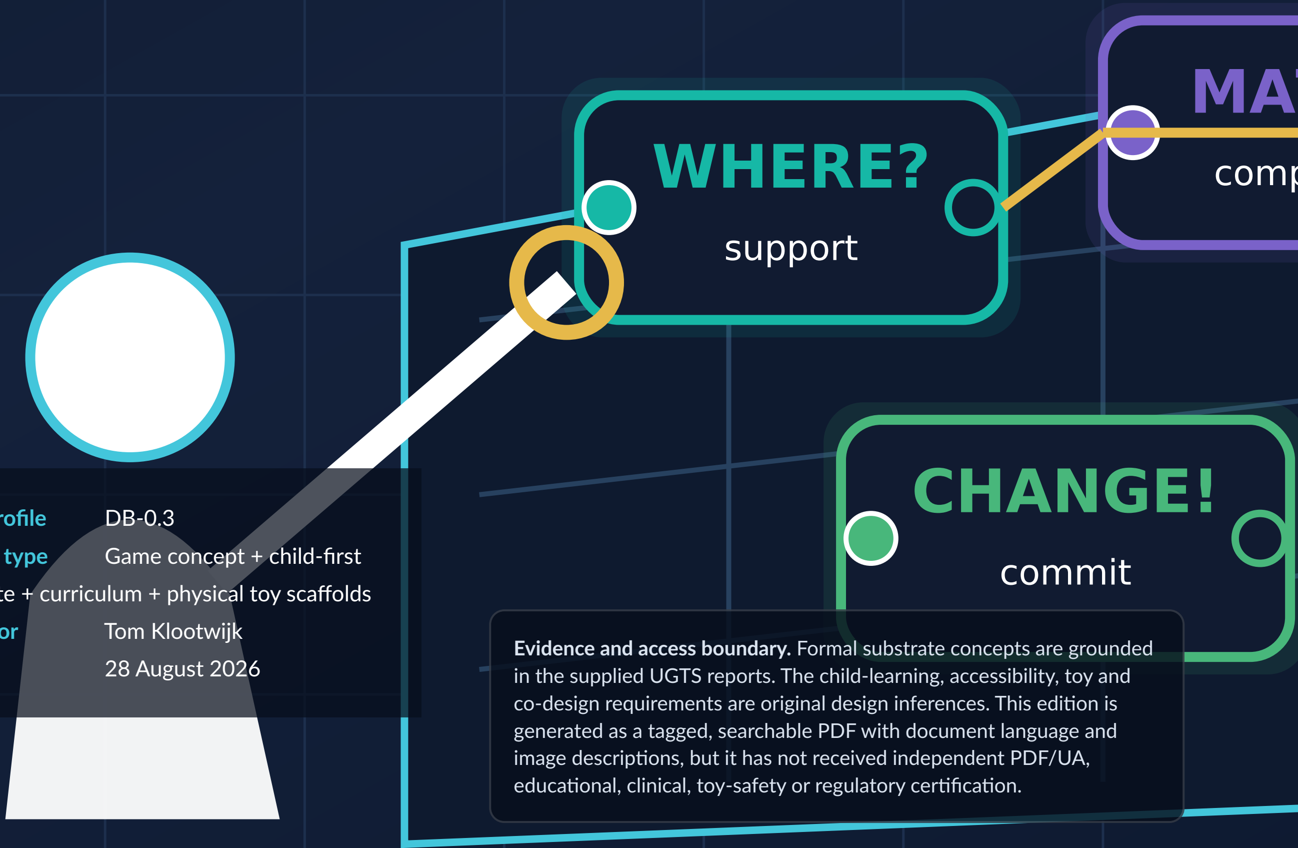
CHILD-FIRST ACCESSIBILITY EDITION

UGTS-KC GAME + LEARNING DESIGN PROFILE

DEFINITION BREAKER

Young Breakers: an accessible puzzle-adventure, learning system and physical-to-digital play scaffold

See it. Hear it. Touch it. Choose it. Explain it. Commit it.



Concept profile	DB-0.3
Document type	Game concept + child-first access route + curriculum + physical toy scaffolds
Prepared for	Tom Klootwijk
Date	28 August 2026

Evidence and access boundary. Formal substrate concepts are grounded in the supplied UGTS reports. The child-learning, accessibility, toy and co-design requirements are original design inferences. This edition is generated as a tagged, searchable PDF with document language and image descriptions, but it has not received independent PDF/UA, educational, clinical, toy-safety or regulatory certification.

START HERE

You are a Breaker

A world is made from rules. In this game, you can hold the rules, try a change and check what happens.

1

Look

Find the rule that is making the problem.

2

Move one thing

Place a rule, match a socket or change the order.

3

Preview safely

See or hear the idea before it changes the real game world.

4

Commit

Make one checked change. The game remembers what changed.

You are in control.

You may pause, ask, point, listen, use one hand, use a helper, take more time or stop.

THE SIX BIG QUESTIONS

Ask these questions

W Where?

Where can this rule matter?

Grown-up word: support

M Match?

Are these pieces allowed to work together?

Grown-up word: compatibility

N When?

Has the important line really been crossed?

Grown-up word: guard

Y Yes?

Has the change been checked?

Grown-up words: verified event

C Change?

What changes now?

Grown-up word: transition

R Remember?

What should the world remember?

Grown-up word: lineage

The letter, word, border style and spoken question all identify the same idea. Color is extra, never the only clue.

SAFE TO TRY

A preview is not the real change

You can test an idea without breaking the world.

1. TRY
Build a candidate idea.

2. PREVIEW
See, hear or feel what it would do.

3. EXPLAIN
The game tells you the first thing that does not work.

4. COMMIT
Only a checked idea changes the world.

When the game says “not yet”

Not here.
Move the rule into the right support zone.

These do not match.
Find the socket, tag or sheet that belongs.

The line was not crossed.
Change the threshold, direction or timing.

Two changes disagree.
Choose an order, priority or merge rule.

A piece is missing.
Repair the reference or dependency.

This would break a promise.
Keep the invariant and try a different route.

“Not yet” is information.
It is not a punishment, a loud alarm or a loss of progress.

YOUR ACCESS MENU

Choose how you play

These choices are available before the first puzzle and can change at any time.

See
Large words, high contrast, thick lines, low glow, no color-only clues.

Hear
Captions, visual sound cues, narration choices, separate music and effects.

Move
One hand, toggle instead of hold, snap-to-socket, no fast dragging.

Control
Keyboard, gamepad, touch, switch-style step selection or a helper.

Pace
Full pause, no timer, slower animation, replay and unlimited safe previews.

Words
Picture first, plain words, spoken help, formal words only when wanted.

Sound and motion
Quiet mode, reduced motion, no surprise effects, intensity preview.

Help
One clue at a time, show the first failed question, co-player mode.

A difficulty choice never removes accessibility choices. Guided, Standard and Expert play can all use the same access tools.

COMMUNICATION COUNTS

There are many ways to answer

A good idea counts even when it is not spoken, written or made quickly.

You can...

- [ ] say the answer
- [ ] point to a card
- [ ] place a rule block
- [ ] choose yes or no
- [ ] use a gesture
- [ ] use a switch or button
- [ ] draw the route
- [ ] ask a helper to move it

A helper can...

- hold or move a piece while you choose;
- read the words without giving the solution;
- repeat your choice back for checking;
- pause immediately when you signal;
- never replace your decision with theirs.

Your choice stays yours.

Communication support helps the child act on an idea; it does not take authority away from the child.

MY ACCESS PASSPORT

What helps me play?

I want...

- [ ] big words
- [ ] pictures with words
- [ ] high contrast
- [ ] less glow
- [ ] quiet sounds
- [ ] captions
- [ ] spoken instructions
- [ ] less movement
- [ ] one-hand controls
- [ ] toggle, not hold
- [ ] more time
- [ ] one clue at a time
- [ ] a co-player
- [ ] a tactile version

My stop signal

I will say, point, press or show:

____

My restart signal

I am ready again when I:

____

The passport records preferences, not diagnoses. The child may change any choice at any time.

PLAY TOGETHER

Every role matters

Observer
Finds the rule, zone or change.

Builder
Moves the pieces chosen by the group.

Checker
Asks the six big questions.

Storyteller
Shows or tells what changed.

Replay Keeper
Puts the change cards in order.

Comfort Keeper
Watches pace, sound, motion and stop signals.

The grown-up promise

1. Ask before helping or touching a child’s pieces.
2. Honor the stop signal immediately.
3. Offer one idea at a time and enough processing time.
4. Do not require speech, eye contact, fast movement or reading aloud.
5. Celebrate prediction, explanation and recovery - not speed.
6. Let roles change. No child is trapped in the “easy” role.

These pages are a child-facing design route. They do not replace individual access planning, professional advice or participatory testing.

HIGH-LEVEL PRODUCT DECISION

Executive Decision

Make the substrate's most unusual capability both the player's main verb and an accessible learning language children can see, hear, touch, point to and control at their own pace.

Definitions, operation order, compatibility, guards and transitions become carryable, connectable world objects. The player can propose radical changes, but the world changes only after a deterministic verification and commit.

DECISION

GO for accessibility as a core system, not a late settings screen.

Every essential idea must work through more than one sense, more than one input method, plain language before formal language, and a safe preview before commit.

GENRE
Systemic puzzle-adventure with light real-time hazards

PRIMARY FORMAT
Tagged offline PDF + accessible HTML5 game + printable and physical-learning prototypes

SIGNATURE MECHANIC
Physically rewire typed rule objects and commit verified world patches

LEARNING SPAN
Ages 5-16+ with easy-read, sensory, motor, communication and formal learning routes

MVP PROMISE
A 7-room digital slice, an easy-read child route and three physical activities with access variants

Why it is cool

Instead of finding a key, the player can change what “key,” “door,” “open,” or “before” means - while the game visibly proves whether the edit is legal.

Why it is playable

The player manipulates a bounded subgraph with typed sockets, budgets, preview and readable rejection reasons. It is tactile puzzle solving, not arbitrary coding.

Why UGTS fits

The source line already separates definitions, instances, support, compatibility, guards, commits, patches and lineage, and 3.9 supplies the practical game shell.

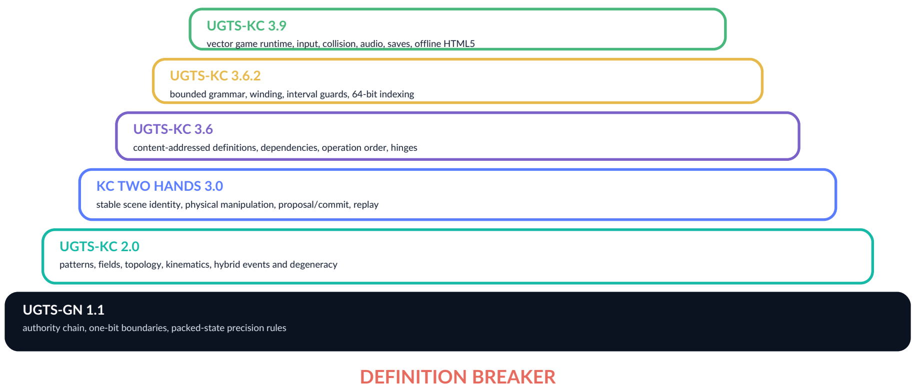
Source-supported foundation. Content-addressed definitions and explicit operation order come from 3.6; event-authoritative proposal/commit and replay from Two Hands 3.0; vector gameplay, input, collision, audio and offline export from 3.9.
S3 pp. 5-13 S4 pp. 3-13 S6 pp. 3-15

SOURCE SCREEN AND EVIDENCE DISCIPLINE

Suitable Substrate Versions

The upgraded concept uses every suitable supplied version according to its documented strength. Core sources support the first playable and learning pathway; the security-specific SARA profile is an optional older-teen boundary lesson rather than a game mechanic.

Source	Status	Contribution to Definition Breaker	Design boundary
UGTS-GN 1.1	CORE	Canonical support -> compatibility -> guard -> event -> transition -> lineage order; narrow one-bit roles; identity and precision boundaries.	This supplies the six-question backbone. GPU/hardware measurements remain engineering evidence, not child-facing claims.
UGTS-KC 2.0	CORE	Patterns, fields, route identity, winding, kinematics, bounded dynamics, event degeneracy and equal-time conflict concepts.	Topology is explicit routing, not physical magic. Numerical status and uncertainty stay visible.
KC Two Hands 3.0	CORE	Stable scene identity, physical manipulation, proposal verification, deterministic commit, checkpoints, replay and reason codes.	Native VR, general physics and live networking are later targets, not prerequisites.
UGTS-KC 3.6	CORE	Definitions as first-class data, explicit dependencies, content addresses, operation order, definition queries and typed hinges.	References must resolve; hidden state and undeclared cycles are rejected.
UGTS-KC 3.6.2 SCLP	CORE+	Winding, reflective wraps, interval guards, constraint release, bounded grammar, seeds, finite keys and storage audits.	A key is an address, not an entire world; finite packing never means infinite detail.
UGTS-KC 3.6.3 SARA	OLDER-TEEN	A safe advanced lesson in typing, public/secret separation, authorization guards and evidence boundaries.	No wallet attack, secret search, signing, transaction or ownership claim belongs in the game or youth activities.
UGTS-KC 3.9	CORE	Vector art, multi-device input, deterministic game world, collision, animation, tilemaps, procedural audio, saves and offline HTML5.	Native 3D, skeletal animation, multiplayer and general rigid-body physics are not claimed.
Definition Breaker DB-0.1	BASELINE	Original 28-page game concept: rule objects, Commit Bench, Rule Lens, puzzle grammar, rooms, campaign and runtime plan.	DB-0.3 preserves that game design and adds the complete child-learning and physical-product pathway.



WORLD, ROLE AND PROMISE

High Concept and Player Fantasy

The premise

The player is a **Breaker**, a maintenance operative inside the Definition Stack: a city whose architecture is compiled from visible rules. After an event called **The Unresolved Cycle**, definitions detach from their instances. Doors inherit gravity, corridors forget which sheet they occupy, alarms and bridges commit contradictory patches, and some rooms depend on themselves.

The Breaker carries a Rule Lens and a portable Commit Bench. They can pull authorized rule objects from sockets, route typed cables, reorder phases, insert hinges, change topology ports and remove constraints. But raw manipulation is only a proposal. The world accepts a change after the substrate proves the edit is well-typed, supported, compatible, numerically safe, invariant-preserving and conflict-resolved.

> "You are allowed to break the rule only after you understand exactly what the rule is."

Player fantasy

- See the hidden logic that makes a world behave.
- Touch and rearrange that logic as tangible machinery.
- Use rejection, replay and lineage as investigative tools.
- Turn an impossible room into a valid new world - without cheating the authority model.

UNIQUE SELLING PROPOSITION

The laws of the level are inventory items.

Definitions are not abstract menu entries. They have mass, sockets, phase labels, dependencies, seals and history. Carrying a rule from one machine to another can be as meaningful as moving a crate.

Not a coding game

Players do not write arbitrary text. They manipulate a constrained, visual and tactile rule graph with authored choices, typed connectors, budgets, previews and undo.

Not pure puzzle-menu play

The changed definitions immediately alter traversal, hazards, NPC behavior, topology, timing and object ownership in the physical room.

Not consequence-free reality editing

Every accepted change leaves a lineage record. Later rooms may inspect what the player changed, in which order, and under which policy.

Game-design inference. The narrative, Breaker role and city are original design choices. Their mechanical credibility comes from the substrate's separation of definitions, instances, verified events, patches and lineage. S4 pp. 3-6 S3 pp. 6, 11-13

THE GAME IN ONE CYCLE

Player Verbs and Core Loop

Inspect

Reveal stable IDs, type signatures, dependencies, operation phase, compatibility tags, guards, invariants and lineage.

Manipulate

Carry, socket, cable, rotate, reorder, branch, gate, hinge, wrap, lock, release or quarantine a bounded rule object.

Reason

Scrub a deterministic preview. Compare current and proposed state. Read rejection reasons as clues.

Exploit

Commit the valid patch, traverse the altered room, evade hazards, recover definitions and preserve the needed lineage.

Moment-to-moment rhythm

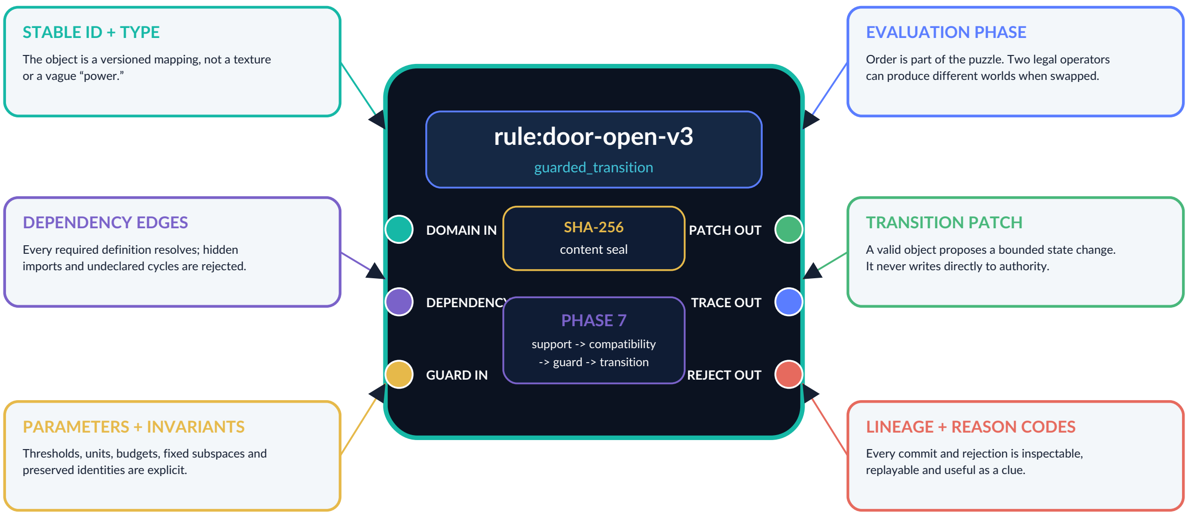
1. Enter a room and discover the physical symptom.
2. Use the Rule Lens to expose the relevant subgraph.
3. Find or earn access to a small set of rule objects.
4. Build a candidate edit on the Commit Bench.
5. Preview consequences and inspect trace/ rejection.
6. Commit, then execute the traversal or timing challenge.

What produces mastery

- Recognizing type and dependency patterns.
- Predicting non-commuting operation order.
- Distinguishing coordinates from identity.
- Understanding crossing, touch, tangency and persistent constraints.
- Planning around deterministic conflict resolution and replay.

DATA MODEL TRANSLATED INTO GAME LANGUAGE

A Rule Object Is a Physical Definition



Non-negotiable design rule. The physical prop is a presentation of a canonical definition record. Picking it up does not change its identity; editing its parameters creates a versioned definition or a patch. A visual appearance, cable or icon never silently becomes the authoritative semantics. S4 pp. 3-7 S3 pp. 5-8

WHAT THE PLAYER CAN CARRY

Rule Object Taxonomy

Literal Tile

Number, vector, tag, threshold, duration, phase, sheet or policy value. Small, stackable and easy to duplicate when allowed.

Typical puzzle: choose the right threshold without shrinking the event margin below safety.

Operator Core

A typed mapping with domain, codomain, parameters and capabilities: transform, compare, combine, project, integrate, route.

Typical puzzle: substitute an operator whose output actually matches the next socket.

Hinge

Continuous pivot, discrete parity selector or zero-magnitude connector with nonzero structural role.

Typical puzzle: insert a connector that changes graph order without changing numeric value.

Support Volume

Local admission region such as sphere, cone, shell, tile region or finite field support.

Typical puzzle: prevent a global rule from affecting unrelated objects.

Compatibility Seal

Sheet, phase, mode, orientation, ownership and application tags that determine whether supported states may couple.

Typical puzzle: make two co-located systems genuinely compatible.

Guard Blade

Boundary, threshold or relation with accepted statuses, confidence floor, error interval and hysteresis.

Typical puzzle: distinguish a true crossing from a tangency or chatter.

Transition Key

An atomic state, route, topology or dynamic reset applied only after verification.

Typical puzzle: change the requested patch without breaking stable node identity.

Topology Port

Explicit gluing, wrap, orientation transport, sheet update and winding lineage.

Typical puzzle: select a half-turn bundle or a genuinely reflective quotient.

Schedule Rail

Stores the actual evaluation/composition order. Blocks on the rail do not commute unless their definitions say so.

Typical puzzle: reorder translate, rotate, gate and commit.

Constraint Shackle

A maintained relation over an interval; removing one declared row may add a legal degree of freedom.

Typical puzzle: release exactly one motion without turning contact into infinite events.

Lineage Seal

Pre/post state hashes, committed event, source, branch and history reference.

Typical puzzle: preserve a required route history while changing position.

Quarantine Cage

Contains unresolved references, invalid hashes, forbidden cycles, overflowed grammar or incompatible definitions.

Typical puzzle: repair the graph rather than bypass validation.

Presentation inference. Names such as “Guard Blade” and “Lineage Seal” are game-language skins over source-supported record classes; the underlying semantics stay typed and inspectable.

MAKE FORMAL STATE READABLE AND TACTILE

The Commit Bench and Rule Lens

Rule Lens: understand the current world

RULE LENS / TARGET: bridge_17

Instance	node:bridge_17
Definition	geom:hinged_bridge-v2
Phase	4 transform
Support	room_03 only
Conflict	translate precedes rotate

```
parse -> resolve -> construct ->
transform -> support ->
compatibility -> guard ->
transition -> lineage
```

The lens has three layers: **simple** (icons and plain language), **trace** (ordered operations and reasons) and **definition** (full typed record for expert players and workshop authors).

Physical input

- **Mouse/touch:** drag blocks, cable sockets, scrub preview, long-press to inspect.
- **Gamepad:** radial selection, snap-to-socket, shoulder-button phase movement.
- **Two-hand future:** stretch the bench, align ports and twist a rule cluster using the 3.0 bimanual transform.

Commit Bench: edit a candidate world

CANDIDATE PATCH / NOT AUTHORITATIVE

3 CONSTRUCT	bridge
4 TRANSFORM	rotate 90 -> translate +2
5 SUPPORT	room_03
6 COMPAT	player + bridge
7 GUARD	angle >= 90
PREVIEW HASH: 3bd1...	VERIFY COMMIT

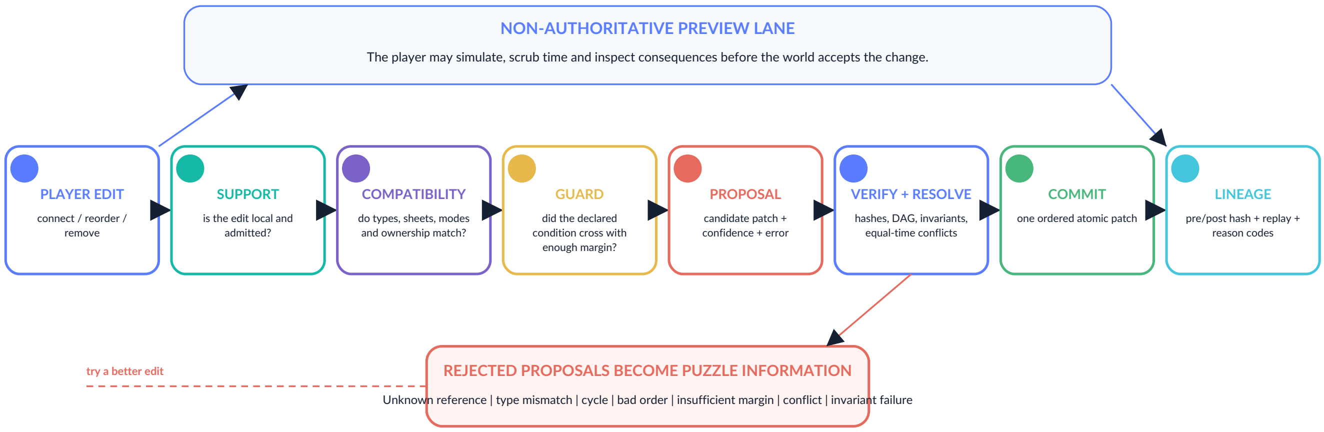
Usability protections

- Only sockets relevant to the current puzzle are exposed by default.
- Every block has shape, icon and text coding; color is never the only cue.
- Invalid cables refuse to snap and explain the domain/codomain mismatch.
- Preview is visually ghosted; the committed world is never ambiguous.
- Undo is checkpoint/replay, not a fragile reverse mutation.
- A hint can reveal one missing gate or one conflicting dependency without solving the entire graph.

Interaction boundary. The 3.9 action map supports keyboard, gamepad, pointer and touch. The richer two-hand manipulation is a later adapter target; the first release should not depend on native OpenXR. S6 p. 6 S3 pp. 11-12, 17

THE GAME'S GOVERNING LAW

Every Edit Is a Proposal, Not a Cheat



Verification checklist

1. Definition IDs resolve and content hashes verify.
2. Dependency graph is acyclic under the selected cycle policy.
3. Socket domains/codomains and units match.
4. Edit lies inside an admitted support zone.
5. Sheet, phase, mode, orientation and ownership are compatible.
6. Guard status is accepted; numerical error is within margin.
7. Declared invariants and stable identities survive.
8. Same-time patch conflicts are ordered, merged or rejected.

Why this is fun rather than bureaucracy

The verifier is the puzzle's "physics." A rejection is concrete information: the player learns that a route is on the wrong sheet, a dependency is cyclic, a crossing is only a touch, or two valid edits conflict at the same tick.

Later levels deliberately require using the verifier as an instrument: construct a failed proposal to reveal a hidden phase, then repair it.

The pipeline preserves the recurring UGTS authority chain and the 3.0 proposal/commit split. S1 pp. 2-4, 28 S2 pp. 11-12, 15 S3 pp. 5, 13

FAILURE BECOMES LEGIBLE

Preview, Rejection and Deterministic Explanation

Current world

The last committed state. Solid colors, collision and gameplay authority are active. Its canonical hash anchors saves, replay and puzzle goals.

CURRENT / 8ef1...

Candidate preview

A sandboxed simulation of the proposed graph. Ghosted geometry, predicted routes, contact traces and potential conflicts are visible, but nothing mutates authority.

PREVIEW / 3bd1...

Committed world

The verified patch is applied atomically; lineage and a replay record are appended. The new world receives a fresh state hash.

COMMITTED / a24c...

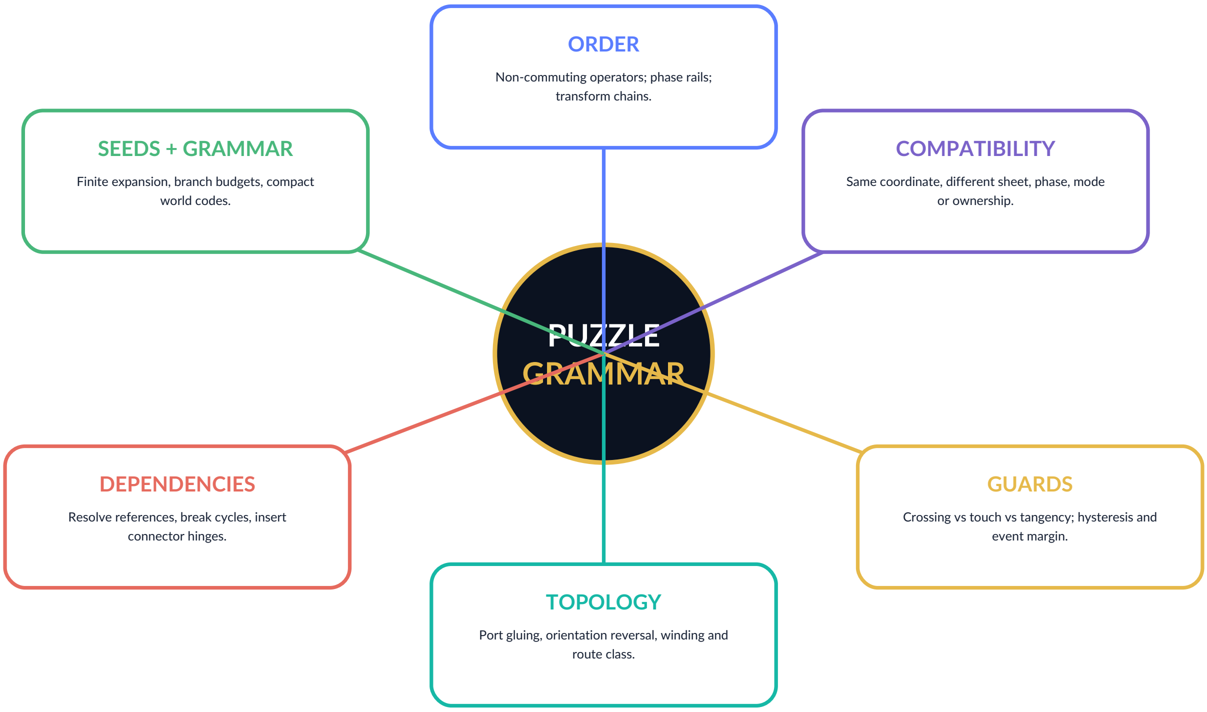
Rejection is a designed feedback channel

Reason code	Player-facing message	Puzzle meaning
`unknown_reference`	“This rule points to something that does not exist here.”	Find the missing definition or replace the dependency.
`type_mismatch`	“A route is not a velocity.”	Insert a projector/adapter or use the correct socket.
`dependency_cycle`	“The door waits for the power that waits for the door.”	Break the cycle with a latch, initial literal or connector hinge.
`incompatible_state`	“These objects overlap, but they do not occupy the same state sector.”	Resolve sheet, phase, mode, orientation or ownership.
`guard_ambiguous`	“The relation interval touches both sides of the boundary.”	Increase margin, improve the measurement or choose a safer guard.
`patch_conflict`	“Two valid rules request different values at the same event time.”	Set priority, prove commutativity or prevent one proposal.
`invariant_failure`	“The edit would erase a stable identity or break topology.”	Change the patch, not the identity; preserve required incidence/gluing.
`budget_overflow`	“This grammar would grow beyond the room's declared limit.”	Reduce depth/branching or reuse a shared definition.

Determinism boundary. The design requires stable ordering and hash-checked replay in the selected runtime profile. It does not assume universal bit-identical floating-point behavior across every future CPU/GPU. S3 p. 13

REUSABLE FAMILIES FOR A FULL CAMPAIGN

The Six-Family Puzzle Grammar



Authored freedom, not arbitrary freedom

Each room exposes a deliberately small editable subgraph: usually 3-8 movable rule objects, 2-5 fixed definitions and one explicit goal. The space is rich enough for multiple valid solutions but bounded enough to read and test.

Solutions are proofs

A solution is not merely reaching the exit. The final state must satisfy the room goal, preserve required invariants and reproduce under deterministic replay from the room checkpoint.

THE FIRST HALF OF THE GAME'S LANGUAGE

Puzzle Families I-II: Order and Compatibility

I. ORDER / NON-COMMUTATIVITY

Two individually valid operators can produce different geometry or state when composed in a different order. The player physically moves blocks along a phase rail or rotates a transform chain.

Core patterns

- Rotate then translate versus translate then rotate.
- Apply a field morph before or after support clipping.
- Evaluate a hinge before a topology wrap so chirality is transported correctly.
- Resolve a guard before a dynamic reset rather than after it.
- Place a connector hinge that changes graph order while carrying zero numeric magnitude.

Readable cues

Ghost paths, phase-colored trails, before/after pivots, and a "first divergence" marker show where two traces become different.

> **Mastery challenge:** construct two different valid orders that reach the same room goal but leave different lineage, then choose the one required by a later audit door.

II. SUPPORT / COMPATIBILITY

Spatial overlap is only the first question. Objects may still be separated by sheet, phase, route class, orientation, mode, policy or ownership.

Core patterns

- Two corridors cross in projection but occupy different sheets.
- A sensor sees the crate, but its support cone excludes the correct face.
- A hazard and player share a tile but are incompatible during a phase window.
- A rule is valid only inside a declared room support, preventing global side effects.
- A two-hand object remains owned after handover even though its position is unchanged.

Readable cues

The Rule Lens separates state sectors into layers, shows support volumes as translucent geometry, and marks incompatible overlaps with a dashed "no coupling" seam.

> **Mastery challenge:** preserve two distinct states at one coordinate until a short compatibility window allows a controlled exchange.

Order as literal data and non-commuting hinges: S4 pp. 5-6, 9, 13. Coordinate equality versus identity/compatibility: S1 pp. 4-5, 18-19 S2 pp. 4-5, 8.

TIME AND ROUTE BECOME MANIPULABLE

Puzzle Families III-IV: Guards and Topology

III. GUARD / EVENT TIMING

A value near zero is not automatically a crossing. The player must reason about direction, derivative, interval, hysteresis, priority and whether a relation should become a maintained constraint.

Core patterns

- Crossing, touch, tangency and coincidence produce different outcomes.
- A fast object needs a swept guard rather than a sampled overlap.
- Enter and exit thresholds prevent a switch from chattering.
- A persistent grab/contact must become a constraint mode, not infinite events.
- Two event-time intervals overlap and require priority, merge or conflict.

> **Signature room:** "The Door That Opens Too Late." Move the predictive guard ahead of collision or add a certified swept relation so the bridge commits before the runner arrives.

IV. TOPOLOGY / ROUTE IDENTITY

Portals and wraps are explicit coordinate, orientation, sheet and lineage maps. The player manipulates route classes rather than relying on impossible visual tricks.

Core patterns

- Möbius-style wrap flips orientation after one traversal.
- Reflective Klein gluing is distinct from a simple half-turn bundle.
- Winding count distinguishes repeated phase coordinates.
- A self-crossing path may represent separate branches or route classes.
- Cell-complex edits must preserve incidence invariants.

> **Signature room:** "Klein Freight." Route a marked crate through a reflective wrap, compensate its chirality, and deliver the correct orientation without merging two incompatible sheets.

Event degeneracy and persistent constraints: S2 pp. 11-12 S3 p. 11. Explicit wrapping, winding and chirality: S5 pp. 5-10.

THE ADVANCED PUZZLE LANGUAGE

Puzzle Families V-VI: Dependencies and Seeded Grammar

V. DEFINITION GRAPH / REFERENTIAL CLOSURE

Definitions may depend on other definitions, but every reference must resolve, hashes must verify and the graph must obey its cycle policy. The player repairs the actual logic graph behind broken machines.

Core patterns

- Missing references create orphaned machinery.
- Duplicate IDs produce ambiguous sockets.
- A definition hash fails after an illicit parameter edit.
- A door-power-door dependency creates an unresolved cycle.
- A connector or latch breaks the cycle while preserving value and order.
- Replacing a definition changes every instance that references it; replacing one instance does not.

> **Signature room:** “Gravity Is a Dependency.” Detach a local room's gravity definition without changing the global definition used by the rest of the station.

VI. SEED / BOUNDED GRAMMAR / COMPRESSION

Rooms and challenge variants can be reconstructed from a seed, finite grammar and compact index, while external novelty and committed edits remain logged.

Core patterns

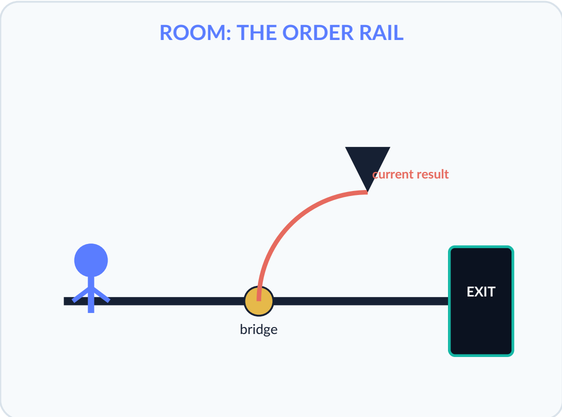
- Branch bits select one of two declared productions.
- Grammar depth, symbols and stack are budgeted.
- A 64-bit address chooses a quantized local cell/phase, not full state.
- Prefix refinement narrows a search region but does not directly halve physical coordinates.
- Seed + definitions reconstruct the baseline; player edits remain event history.

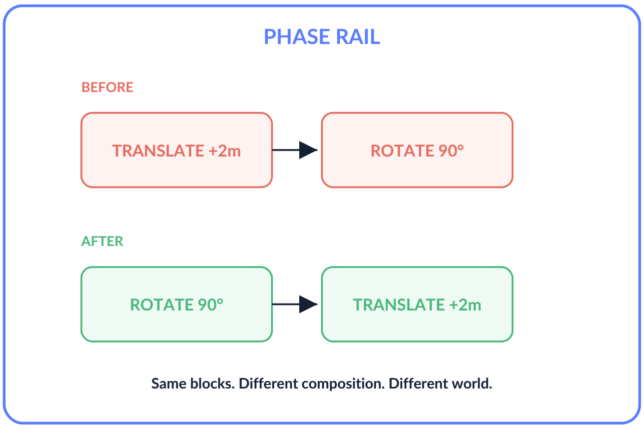
> **Signature room:** “Worldseed 64.” A map key locates the missing rule cache, but the player must recover the separate definition and lineage records to make the room whole.

Compression boundary. The game may exploit shared definitions, vector assets, bounded procedural generation and event-log reconstruction. It must never present nominal bit-width ratios as equal-information compression without a reconstruction and error contract. S5 pp. 7-12 S1 pp. 9-13

DETAILED PLAYABLE SCENARIO 1

Example Room: The Order Rail





Room symptom

A hinged bridge rotates around the wrong point and misses the exit ledge. The same translation and rotation blocks are already present; the machine is “valid” but ordered incorrectly.

Intended discovery

Composition is non-commutative. The Rule Lens marks the first divergence between the current and candidate transform traces.

Solution

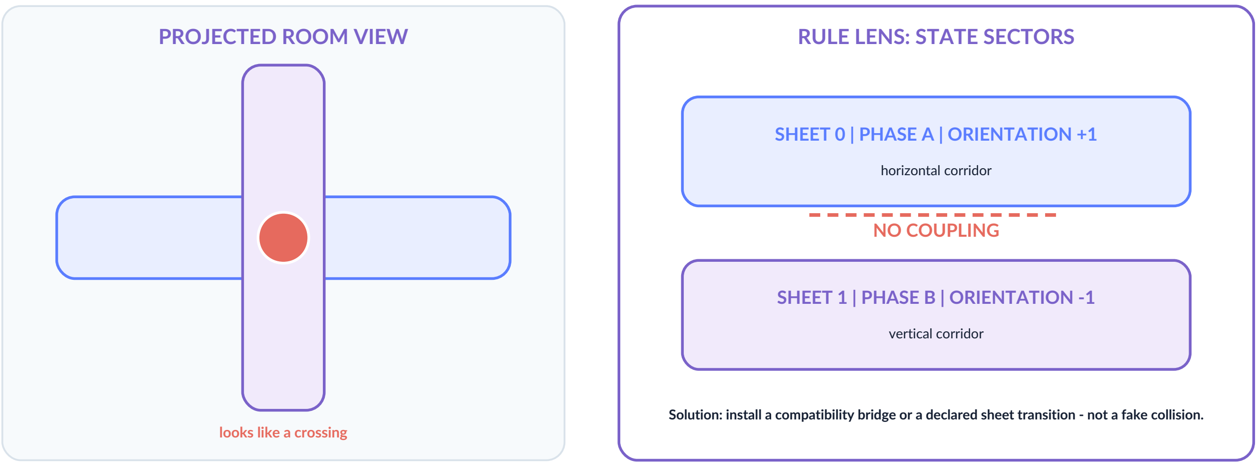
Move **ROTATE 90** before **TRANSLATE +2m** on the phase rail. Preview the pivot path, verify, then commit the transform-order patch.

Available objects	Hidden constraint	Failure feedback	Optional expert solution
Rotate, translate, pivot literal, room support, angle guard.	Bridge node identity and hinge pivot must remain stable.	“Invariant failure: pivot moved with the asset.”	Keep the original order but insert a compensating frame transform; produces a different lineage accepted by the goal.

Teaching purpose. This room turns 3.6's explicit non-commuting composition and 3.0's stable node identity into a readable first-act puzzle. S4 p. 6 S3 p. 6

DETAILED PLAYABLE SCENARIO 2

Example Room: Same Place, Different Sheet



Room symptom

The player sees a corridor cross and expects to turn upward, but the avatar passes underneath. A power pulse appears to touch a receiver yet never couples.

Available edits

- Compatibility bridge: temporarily match sheet and phase.
- Declared sheet transition: move one route to the other sheet.
- Projection collision: tempting but invalid; it treats appearance as authority.

Intended solution

Install a compatibility window gated by the pulse phase, allowing energy to transfer for exactly one tick while the player routes through a separate portal. The world remains layered; the solution does not collapse two identities into one.

Failure feedback

`incompatible_state`: “Supported overlap found. Coupling rejected: sheet 0 / phase A versus sheet 1 / phase B.”

Conceptual boundary. The game may visualize sheets as layered corridors, but it should not imply that topology creates physical energy or that projected self-intersection is automatically contact. S1 pp. 4, 18-19 S2 p. 8

DETAILED PLAYABLE SCENARIOS 3-4

Example Rooms: Missing Shackle and Klein Freight

THE MISSING SHACKLE

A suspended counterweight is constrained in two velocity directions. The exit mechanism needs one new legal degree of freedom, but simply deleting all constraints drops the mass and destroys the timing sequence.

Available objects

- Two constraint rows, one damped hinge, tangent projector, safety guard and reset.
- A “release one row” tool that exposes the nullity change.

Solution structure

1. Remove exactly the lateral constraint row.
2. Keep the radial row and tangent projector.
3. Attach the damped hinge so the arm swings rather than accelerates without bound.
4. Commit only when the safety guard certifies the new path.

Core lesson: constraint release increases legal motion; it does not automatically supply force, chaos or a trajectory.

KLEIN FREIGHT

A cargo crate must traverse a radial wrap. The displayed path looks like a half-turn, but the destination scanner requires a genuinely orientation-reversed state and a preserved winding label.

Available objects

- Source half-turn bundle, reflective Klein wrap, chirality automorphism, orientation seal and winding counter.

Solution structure

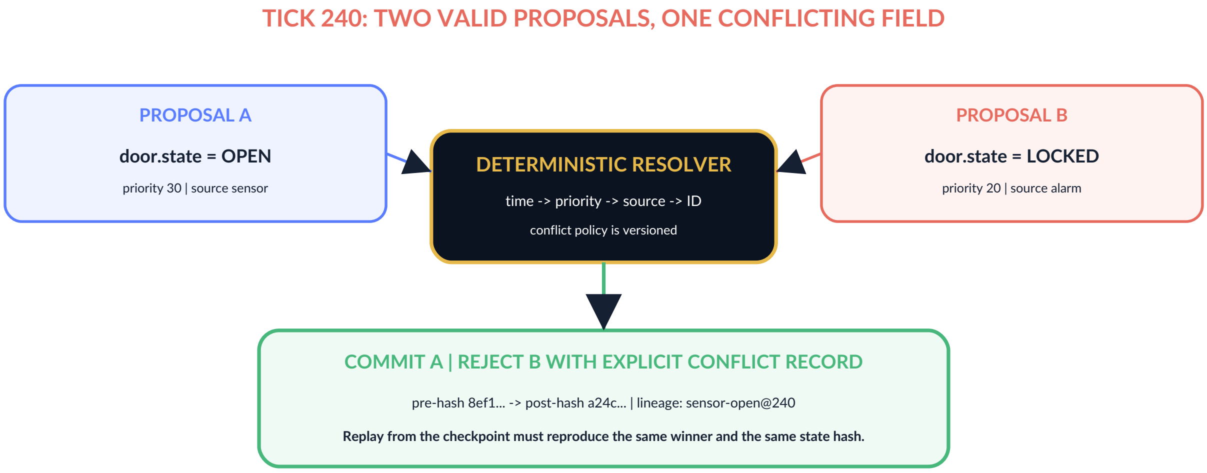
1. Select the reflective wrap rather than the orientation-preserving base rotation.
2. Let the wrap flip local orientation and hoop state.
3. Apply the chirality map to the crate's control convention.
4. Preserve the winding in lineage so the scanner distinguishes first and third arrival.

Core lesson: visually similar wraps can have different derivative/orientation maps and therefore different gameplay semantics.

Constraint row release and tangent projection: S5 pp. 6-7. Half-turn versus reflective Klein profile and chirality: S5 pp. 5-10.

DETAILED PLAYABLE SCENARIO 5

Example Room: Conflict at Tick 240



Room setup

A sensor proposes **OPEN** while an alarm proposes **LOCKED** at the same fixed tick. Both proposals independently satisfy support, compatibility and guard checks. The door cannot accept both values.

Player task

Trace the previous loop, discover why the alarm fires, then either prevent one proposal, change the declared priority, or replace the patches with a commutative state representation such as an explicit *open-but-alarmed* mode.

Replay reveal

The game rewinds to the checkpoint and replays input and events. A ghost trace shows the same conflict and the same winner. The player can scrub directly to the first hash divergence after an edit.

Expert variant

Two players or two echoes each control one source. They must coordinate timing so the proposals no longer overlap, avoiding a priority hack.

Design inference. Turning deterministic conflict resolution into a time-loop puzzle is an application concept. The underlying ordered proposal, rejection, pre/post hash and replay mechanisms are source-supported. S2 pp. 11-12, 21 S3 p. 13 S6 pp. 6, 9, 12

TEACH ONE DEEP DISTINCTION AT A TIME

Campaign Structure and Progression

The campaign increases expressive power, but never removes typing, budgets, preview or deterministic commit.

Chapter	Rule vocabulary	Mechanical escalation	Boss / set-piece
I. The First Hash	Stable ID, literal, type, instance versus definition.	Match sockets; repair one missing reference; learn preview/commit.	**The Orphan** - a machine with no definition.
II. The Phase Rail	Operation order, hinges, transforms, invariants.	Reorder blocks; preserve pivots; exploit connector hinges.	**The Commutator** - changes order every loop.
III. False Twins	Support, compatibility, sheet, phase, ownership.	Manage co-location without false coupling; timed compatibility windows.	**The False Twin** - two identical bodies with different lineage.
IV. The Event Horizon	Guard margin, crossing, touch, tangency, hysteresis, constraint.	Predict events; avoid chatter; convert persistent contact to a maintained mode.	**The Chatter Engine** - floods the room with invalid zero events.
V. Winding City	Port gluing, orientation, sheet change, winding and route class.	Route cargo, player and signals through different quotient rules.	**The One-Sided Station** - orientation is the actual health bar.
VI. Replay Court	Proposals, priorities, conflict, checkpoints, hashes, lineage.	Use past loops as proof; construct reproducible multi-event solutions.	**The Auditor** - accepts only solutions whose replay and lineage match.
VII. Definition Breaker	Dependencies, bounded grammar, seed/index, full authority chain.	Repair the Unresolved Cycle and decide which definitions the city should keep.	**The Optimizer** - tries to collapse identity, uncertainty and history into one bit.

TURN FORMAL DISTINCTIONS INTO MEMORABLE DRAMA

Narrative, Characters and Boss Logic

The thematic conflict

Definition Breaker is about the difference between freedom and unverified power. The city failed because its administrator, **The Optimizer**, attempted to compress every distinction - identity, state, uncertainty, route history and meaning - into minimal flags. The resulting "efficient" world is unable to explain itself and cannot safely change.

The Breaker restores not a single correct order, but an inspectable order: typed definitions, explicit dependencies, bounded edits and preserved lineage. The final choice can offer multiple valid city constitutions, each replayable and auditable.

Key characters

- **Mira / The Instance:** a recurring NPC whose body and asset change, but whose stable node identity and lineage persist. Teaches identity separation.
- **Latch:** a small connector-hinge companion that has "zero magnitude, nonzero structural role." It becomes a literal puzzle tool and comic voice.
- **The Auditor:** antagonist-turned-ally who refuses undocumented miracles; teaches proof, replay and reason codes.
- **The Orphans:** machines and citizens whose definitions are missing. Restoring them raises ethical questions about version and provenance.

THE COMMUTATOR

Reorders transform and event blocks. Damage is dealt by pinning a deterministic phase order long enough to make its own attacks miss.

THE FALSE TWIN

Creates co-located duplicates on incompatible sheets. The player must target lineage and compatibility rather than appearance.

THE CHATTER ENGINE

Oscillates around a guard boundary. The solution is hysteresis, hold time or a constraint mode - not faster button pressing.

THE ONE-BIT KING

Claims one selector contains geometry, state and identity. The arena is won by splitting those roles into explicit objects.

THE OPTIMIZER

Deletes uncertainty and lineage to make every patch "fast." The final fight uses evidence, conflict traces and reconstruction to prove why a smaller record can be less truthful.

Narrative inference. The characters and themes are original. They dramatize explicit source boundaries: one-bit narrow roles, coordinates not identity, packed hashes not complete lineage, and compression not correctness. S1 pp. 10, 13, 15 S5 pp. 11-14

MAKE LOGIC INSTANTLY READABLE

Art Direction, Animation and Procedural Audio

Visual language

support compatibility guard commit rejection trace

- **World:** clean vector architecture, dark substrate voids, luminous path networks and readable silhouettes.
- **Rule objects:** semi-transparent blocks with physical sockets, phase bands, content seals and animated dependency cables.
- **Preview:** ghost geometry, dotted collision volumes and alternate route overlays.
- **Topology:** sheet separation shown through parallax/ layer offset rather than impossible camera tricks alone.
- **Definitions:** resolution-independent SVG/Canvas paths; no texture atlas required for the first slice.
- **Patterns:** Gielis, Lissajous, rose, spiral and periodic-field contours give machines distinct identities without sampled art.

Animation

Rule blocks pulse by state: idle, candidate, verified, rejected and committed. Keyframe clips, easing, ping-pong clocks, crossfades and property-path application are enough for the first release.

Audio grammar

Event	Sound logic
Support admitted	Short low click; one oscillator voice.
Compatibility complete	Second interval joins the first; consonance indicates matched tags.
Guard approaching	Pitch rises with normalized residual, but never pretends to be probability.
Verified proposal	Three-note cadence; still unresolved until commit.
Commit	Full chord plus percussive transient; lineage motif appended to music.
Rejection	Dissonant interval shaped by reason family, followed by a readable text cue.
Replay divergence	Beat sequence splits at the first mismatched hash.

AUDIO TARGET

The player should hear the authority chain.

Procedural oscillator/noise cues and beat sequences keep the package compact and make each verification stage musically legible.

Screen output

The first game can use the 3.9 Canvas renderer, gradients, transforms, z-order, camera, grid, vignette, particles and HUD. More advanced field visualization remains a later extension and must preserve the same authority boundary.

Vector, animation and procedural-audio support: S6 pp. 5-7, 10, 14. Pattern families: S2 pp. 5-7, 16+.

DEPTH WITHOUT EXCLUSION

Accessibility, Difficulty and Hint Design

Perceptual access

- Color + icon + shape + text for every rule class.
- High-contrast mode and reduced glow.
- Adjustable trace thickness and animation speed.
- Screen-reader labels for definition inspector and reason codes.
- No solution depends only on tiny moving particles.

Motor access

- Snap-to-socket and one-button cable routing.
- Full pause during rule editing in standard mode.
- Hold/toggle options for Rule Lens.
- Remappable named actions across keyboard, gamepad and touch.
- Undo to checkpoint without timing precision.

Cognitive access

- Plain-language layer before formal type detail.
- One new distinction per early room.
- Trace highlights the first divergence, not every intermediate value.
- Reason codes grouped into reference, type, compatibility, guard, conflict and invariant families.
- Glossary terms unlock in context.

Three difficulty profiles

Profile	Editing time	Inspector	Failure assistance	Best for
Guided	World pauses fully.	Plain language, relevant sockets only.	Highlights the first failed gate and suggests one next action.	Story players and accessibility.
Standard	World pauses at benches; hazards continue in late chapters.	Plain + trace layers.	Full reason code and before/after comparison.	Intended first playthrough.
Expert / Audit	Optional live edits and tighter budgets.	Full record, hashes and dependency graph.	No suggested action; deterministic trace remains available.	Optimization, speedruns and workshop validation.

Hint ladder

1. Symptom
2. Relevant phase
3. First failed gate
4. Missing rule class
5. Candidate wiring
6. Full solution

Design goal. The formal substrate should increase explainability, not create jargon walls. The expert layer is optional; the physical world and plain-language rejection should carry the main experience.

ADDITIVE SCHEMA, INSPECTABLE RECORDS

Technical Game Data Model

Project-level additions

The first implementation can extend the 3.9 `GameProject` additively rather than inventing a separate engine:

```
{
  "schema": "ugts-kc-definition-breaker-0.1",
  "rule_library": [...content-addressed definitions...],
  "rule_instances": [...physical blocks and fixed
sockets...],
  "edit_zones": [...what the player may manipulate...],
  "commit_policies": [...accepted statuses, margins,
conflicts...],
  "puzzle_goals": [...state, route, lineage and replay
predicates...],
  "scenes": [...ordinary 3.9 entities, tilemaps, art and
audio...]
}
```

Canonical rule definition

```
{
  "id": "rule:door-open-on-power-v1",
  "kind": "guarded_transition",
  "domain": "door_state x power_state",
  "codomain": "door_state",
  "evaluation_phase": 7,
  "dependencies": ["guard:power-positive-v1"],
  "parameters": {"threshold": 1.0},
  "invariants": ["door_node_id_stable"],
  "provenance": {"class": "game-authored"},
  "content_hash": "sha256:..."
}
```

Runtime records

RuleInstance
id, definition_ref, transform, socket_state, ownership, edit_zone, lineage_ref

RuleSocket
domain, codomain, phase, multiplicity, allowed_kinds, lock_state

RuleEditProposal
id, tick, editor, target, operation, definition_ids, expected_pre_hash, confidence, numeric_error, margin, lineage_label

RulePatch
add/remove/replace/reorder/connect/disconnect/parameter/topology/constraint operation

PuzzleGoal
required state predicates, route class, preserved invariants, forbidden reason codes, replay hash policy

RuleTrace
ordered phase outputs, first divergence, gate results, conflict decisions, pre/post hashes

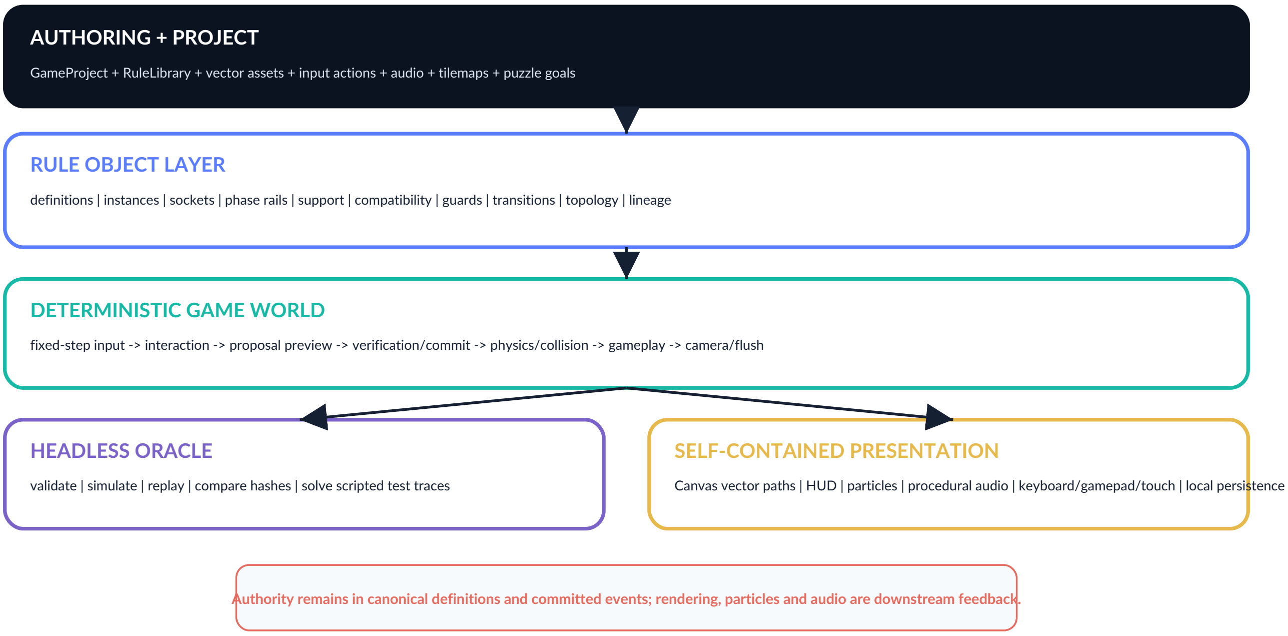
Identity rules

- Definition ID != physical rule-block instance ID.
- Scene-node ID != asset ID != current coordinate.
- Content hash verifies canonical semantics but does not replace lineage.
- Replacing a referenced definition may update many instances; moving one block does not create a new definition.
- Rendered icon and color are downstream views of the record.

This schema is an application-level proposal derived from the 3.6 definition record, 3.0 scene/event records and 3.9 project model.
S4 pp. 3-7, 11-12 S3 pp. 6-7, 19 S6 pp. 3-4

ONE PROJECT, TWO EXECUTION TARGETS

Runtime Architecture and System Order



Suggested components

`Transform2D`, `Body2D`, `Collider2D`, `RuleCarrier`, `DefinitionRef`, `RuleSocket`, `SupportGate`, `CompatibilityTags`, `GuardRelation`, `TransitionPatch`, `TopologyState`, `ConstraintMode`, `LineageRef`, `PuzzleGoal`, `InspectorHint`.

Suggested fixed-step insertion

1. Resolve `InputFrame`.
2. Player controller and rule-object manipulation.
3. Assemble candidate graph and preview requests.
4. Verify/commit accepted rule proposals at a stable pre-physics phase.
5. Physics, bounds and collision.
6. Gameplay hazards, pickups and puzzle-goal checks.
7. Animation, lifetime and procedural feedback.
8. Camera, deferred destruction, snapshot/hash/save.

Implementation decision. Keep the rule system as custom, stably ordered systems inside the existing fixed-step world. Do not let the browser UI mutate entity dictionaries directly. All committed edits go through the same proposal and patch path used by headless tests.

SMALL, DETERMINISTIC AND HONEST

Persistence, Seeds, Compression and Performance

Save/reconstruction strategy

1. Store immutable/vector definitions once and reference them by stable ID/content hash.
2. Generate baseline room geometry from project seed and bounded grammar where reconstruction is deterministic.
3. Store current compact hot state: stable IDs, transforms, modes, phase/sheet/orientation, active constraints and lineage references.
4. Append authoritative player edits, topology changes, conflict decisions and external inputs as events.
5. Create checkpoints when replay cost or divergence risk exceeds the room budget.
6. Verify a loaded save by replaying from the nearest checkpoint and comparing the canonical state hash.

64-bit key use

Use a finite key as a local room/phase index or deterministic challenge code. It may accelerate lookup and prefix refinement, but must not stand in for stable identity, definitions, uncertainty, route state or full lineage.

Compactness advantages

- Vector paths and gradients replace texture atlases for the first release.
- Procedural oscillator/noise cues replace sampled effect files.
- Shared definition records prevent repeated rule payloads.
- Seeded room variation stores parameters and branch choices rather than materialized copies.
- Verified-event compaction avoids dense logging of unchanged candidates.
- Cold definitions/editor metadata stay outside the hot simulation record.

Targets to measure, not claims

Target	Initial design goal
Fixed-step rate	60 ticks/s target on ordinary desktop browsers; measure p50/ p95/p99.
Room rule graph	Usually <= 64 active rule instances and <= 256 definition dependencies.
Grammar	Explicit depth/symbol/stack budgets; reject overflow before simulation.
Preview latency	Immediate for local edits; degrade by reducing horizon, not by hiding status.
Web package	Small enough for instant offline opening; exact byte target set after the first slice.

Precision rule. Packed/fixed-point or reduced records are accepted only when coordinate and guard error remain below the declared event margin and do not change ordering. Memory reduction is not correctness. S1 pp. 9-15 S5 pp. 8-14

SHIP THE SIMPLEST DISTINCTIVE VERSION FIRST

Platform, Modes and Expansion Path

FIRST: OFFLINE HTML5

Desktop browser and static-file distribution using 3.9's Canvas/Web Audio runtime.

- Keyboard, gamepad, pointer and touch.
- Headless Python validation and simulation.
- Single-file or split bundle.
- Local saves and best-solution records.

RECOMMENDED MVP

SECOND: NATIVE DESKTOP / ANDROID

Port the stable record and commit core only after the puzzle schema is proven.

- Native input and storage.
- Optional field/GPU presentation.
- Phone-scale touch Rule Bench.
- Reuse compact vector/seed assets.

AFTER VERTICAL SLICE

LATER: TWO-HAND / CO-OP

Use bimanual transforms and deterministic ownership for spatial rule manipulation.

- Local co-op “dual authority.”
- VR hand joints and handover.
- Server-authoritative workshop sessions.
- Replay and interest management.

NOT MVP

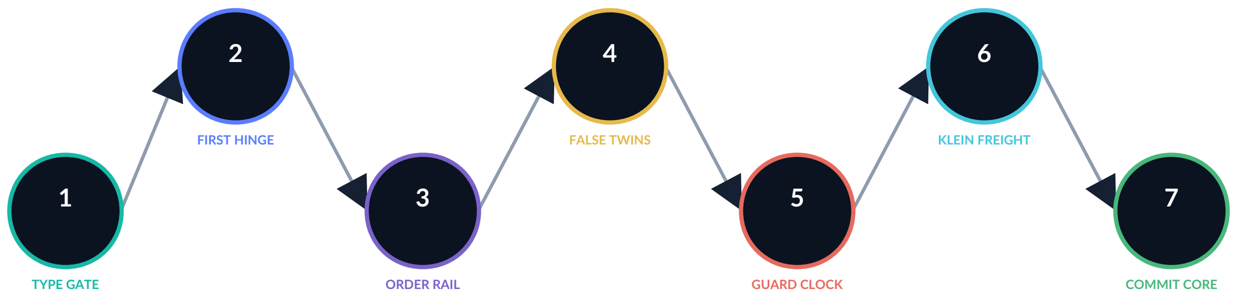
Game modes

Mode	Core promise	Substrate leverage
Campaign	Authored puzzle-adventure with narrative and bosses.	Full rule vocabulary, fixed-step runtime, checkpoints and progression.
Worldseed Rooms	Daily compact challenge code with verified solution replay.	Bounded grammar, 64-bit index, state hash and input/event recording.
Audit Trials	Optimize a room under rule-count, event-count or lineage constraints.	Reason codes, trace metrics, canonical hashes and deterministic conflicts.
Workshop	Author and share bounded puzzle graphs without arbitrary scripts.	Content-addressed definitions, schema validation, source/asset hashes; optional evidence ledger.
Dual Authority	Two players hold different rule classes and must coordinate commits.	Ownership, handover, proposals, same-time conflicts and replay.

Roadmap boundary. The supplied reports do not claim a finished multiplayer service, native OpenXR binding or general console/mobile engine. Those remain explicit later engineering tasks. S3 pp. 15-17 S6 p. 15

THE SMALLEST RELEASE THAT PROVES THE IDEA

Recommended Vertical Slice



VERTICAL SLICE TARGET

20-30 minutes | 7 rooms | 1 rule bench | 1 deterministic replay reveal | 1 topology boss | offline HTML5 build

Playable scope

- One compact facility wing, 7 rooms, 20-30 minutes.
- Player movement, dash, health/hazard fallback and camera from 3.9.
- Rule Lens with simple and trace layers.
- Commit Bench supporting connect, disconnect, reorder and replace.
- Rule objects: literal, operator, hinge, compatibility seal, guard and transition.
- One deterministic replay sequence and one same-tick conflict.
- One orientation-wrap boss using sheet/orientation state.
- Self-contained offline HTML5 build plus headless smoke trace.

Acceptance metrics

Gate	Pass condition
Clarity	New players solve the first three rooms without opening the formal record layer.
Authority	No UI or presentation path mutates canonical state outside verified commit.
Replay	Every accepted room solution reproduces its final state hash from checkpoint.
Reason codes	Every rejected edit reports one primary cause and exact target/ reference path.
Boundedness	No room can create unbounded grammar, event chatter or dependency growth.
Delivery	Project validates, simulates headlessly and opens offline in a browser.
Fun	At least one room produces an “I changed the law” moment rather than a cable-matching chore.

VERTICAL-SLICE DECISION

Prove Definition -> Physical Object -> Preview -> Verified Commit -> Changed Traversal.

Do not begin with a general-purpose visual programming editor. Build seven authored rooms around a narrow rule taxonomy, then expand only after the loop is enjoyable.

BUILD, MEASURE, AND STOP WHEN THE PREMISE STOPS HELPING

Implementation Phases, Kill Criteria and Source Register

Implementation phases

0. Contract	Rule schema, canonicalization, definition hashes, dependency validation, proposal and patch records.
1. Tactile loop	Physical rule blocks, sockets, Rule Lens, preview state and verified commit in one test room.
2. Puzzle proof	Four rooms: type, order, compatibility and guard. Headless replay and reason-code tests.
3. Distinctive content	Topology room, conflict/replay room, boss, procedural audio and final art pass.
4. Productization	Campaign shell, accessibility, saves, challenge codes, build reports, manifests and offline packaging.

Kill or simplify when...

- Players must read raw JSON to understand normal rooms.
- The editable graph grows faster than support/ compatibility can prune it.
- Invalid edits return generic failure rather than a useful reason.
- Visual preview cannot clearly distinguish candidate from committed state.
- Packed precision or cross-platform behavior changes event order.
- Topology is only cosmetic and never alters route, orientation or compatibility.
- Most content is cheaper and clearer as ordinary authored scripting.
- The four-room prototype feels like wiring homework instead of world transformation.

Source register

ID	Source and most-used pages
S1	**UGTS-GN 1.1**, pp. 2-5, 9-15, 18-19: canonical authority, identity, packed records, precision, memory, one-bit and evidence boundaries.
S2	**UGTS-KC 2.0**, pp. 4-15: pattern/field roles, route identity, kinematics, hybrid events, degeneracy, memory and final definition.
S3	**KC Two Hands 3.0**, pp. 5-13, 15-16: scene identity, bimanual interaction, proposal/commit, replay, records and production boundaries.
S4	**UGTS-KC 3.6**, pp. 1-14: content-addressed definitions, dependency graph, hinges, operation order, traces and kill criteria.
S5	**UGTS-KC 3.6.2 SCLP**, pp. 5-15: interval guards, winding, reflective wrapping, constraint release, bounded grammar, keys and packing audits.
S6	**UGTS-KC 3.9**, pp. 3-15: vector art, input, animation, tilemaps, audio, fixed-step game world, persistence and HTML5 export.
S7	**UGTS-KC 3.6.3 SARA**, pp. 1-10, 13-15: typed public/secret boundary, authorization guard, public-only certificates and explicit rejected scopes.
S8	**Definition Breaker DB-0.1**, 28 pages: prior game concept and technical-design baseline expanded by this edition.

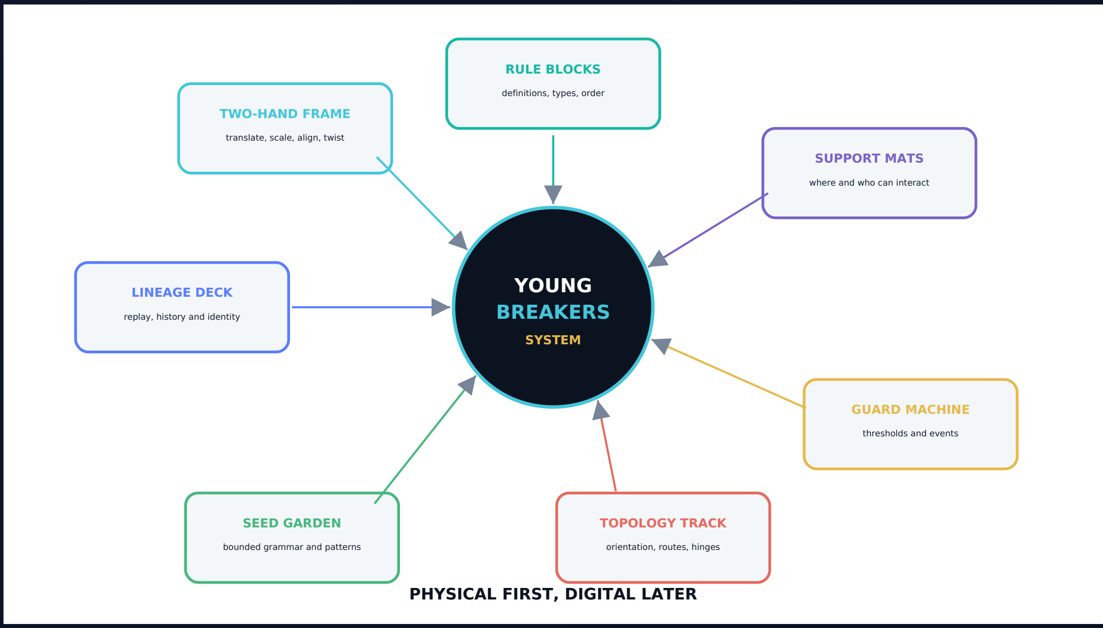
Final evidence boundary for Part I. Formal terminology and implementation boundaries come from the supplied sources. Rooms, characters, interfaces, learning progression and accessibility requirements are application inferences that need prototype evidence.

The strongest version of *Definition Breaker* is not a game where “anything is possible.” It is a game where every impossible-looking change becomes possible only after the player can understand it through an access route that works for them.

PART II / YOUNG BREAKERS EDITION

THE SUBSTRATE AS A CHILD CAN HOLD IT

A layered pathway from story to toy, from tabletop proof to digital rule-breaking, and from intuition to the full formal substrate.



Design boundary. This section translates the supplied substrate into educational experiences. It is not a claim of validated pedagogy, age certification, manufactured toys or completed companion games.

ONE SYSTEM, SEVERAL LEVELS OF PRECISION

Learning Decision and Age-Band Ladder

Children should meet the substrate as a sequence of understandable questions, not as a wall of equations or code.

The learning system preserves the same concepts at every depth. A five-year-old may say “this token only works inside the blue circle.” A teenager may later identify that statement as a support predicate. The idea is not replaced; its vocabulary becomes more precise.

LEARNING DECISION

PHYSICAL FIRST, DIGITAL SECOND, FORMAL THIRD.

Use stories and manipulatives to establish causality, order, identity, thresholds and memory before asking learners to navigate a digital definition graph.

PRIMARY LEARNER PROMISE

“I can explain why the world changed.”

ADULT PROMISE

No coding or mathematics prerequisite for the first layer.

FORMAL PROMISE

Every simple activity maps to a real substrate term.

EVIDENCE PROMISE

Uncertainty, rejection and boundaries remain visible.

PRODUCT PROMISE

All physical products begin as unvalidated prototypes.

AGES 5-7

Story and sorting

Inside/outside, matching, before/after, cause/effect and remembering a sequence.

AGES 8-10

Gates and routes

Support zones, compatibility tags, thresholds, route flips and simple event order.

AGES 11-13

Systems and proofs

Typed sockets, dependencies, non-commutative order, bounded grammar and replay.

AGES 14-16+

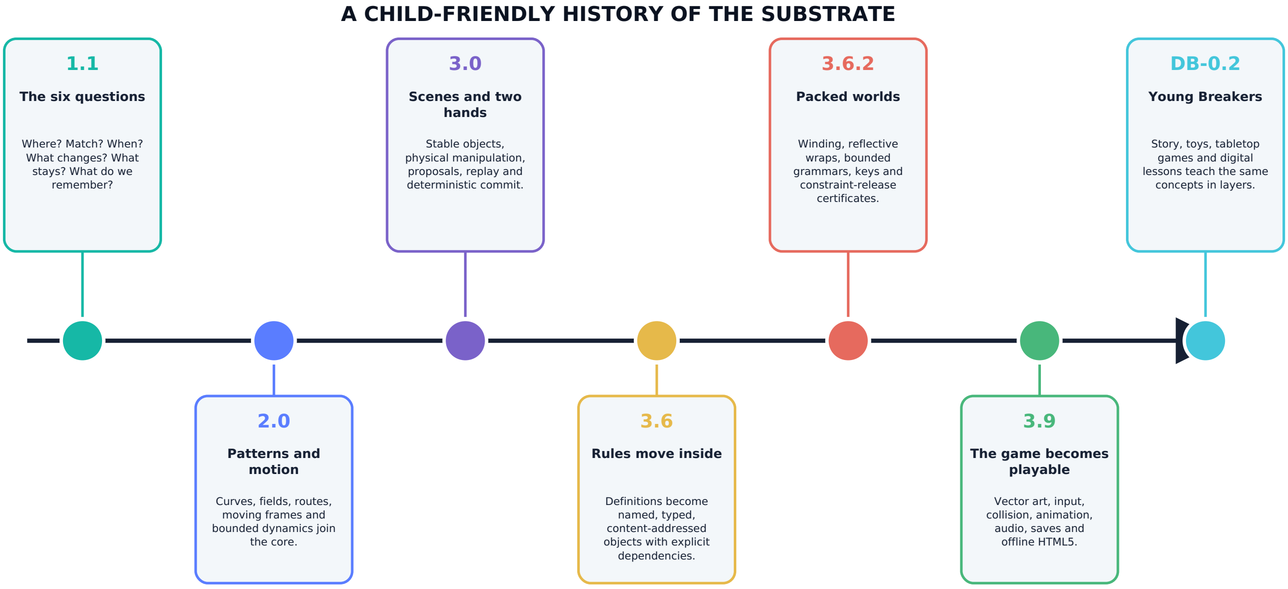
Formal substrate

Definition records, guards/margins, topology maps, compact keys, state hashes and audit boundaries.

Age-band boundary. These bands are design starting points, not ability labels or validated developmental claims. Learners may enter at any layer, revisit earlier manipulatives, or use the formal layer with adult support.

A CHILD-FRIENDLY VERSION LINEAGE

How the Substrate Came to Be



Each version kept the authority chain and added a new way to describe, manipulate or play with it.

The recurring insight

Across the supplied reports, the center stays stable: ask locally where something may matter, whether states may interact, whether a guarded change is valid, what patch occurs, and how identity/history continue.

The growth pattern

Later versions do not discard the earlier questions. They add patterns and motion, physical interaction, definitions inside the data, packed addresses and finally a practical vector-game runtime.

Source-derived lineage: S1 establishes the query-first authority chain; S2 adds patterns, topology, kinematics and bounded dynamics; S3 adds production interaction and replay; S4 makes definitions first-class; S5 adds packed finite profiles; S6 makes the system playable.

THE VOCABULARY LADDER

The Six Questions at the Heart of UGTS

support -> compatibility -> guard -> event -> transition -> lineage

Child language	Formal term	What stays true	First physical activity
Where can this rule matter?	**Support**	Local admission happens before expensive or dangerous interaction.	Place tokens inside and outside a marked zone.
Are these allowed to match?	**Compatibility**	Co-location alone is not enough for coupling.	Match shape/color/tactile tags before connecting pieces.
When is change allowed?	**Guard**	A declared condition and margin decide whether an event is valid.	Cross a threshold line with a token or marble.
Which valid change happened first?	**Event**	Timing and conflict policy are explicit.	Order timestamped cards.
What changes now?	**Transition**	One bounded patch changes the world.	Flip a gate or replace one state tile.
What must we remember?	**Lineage**	Identity and route history survive coordinate changes.	Build a replay strip of cards.

Canonical chain and identity boundary: S1 pp. 2-5 and S2 pp. 4-5, 11-15.

MECHANISM FAMILIES WITHOUT LOSING THEIR BOUNDARIES

The Entire Substrate as a Learning Map

THE COMPLETE SUBSTRATE AS A LEARNING MAP

Different reports add different rooms to one house; the authority chain remains the hallway.

Catalog/profile family	Child-accessible umbrella	Formal coverage
M001-M197 / GN 1.1	The six-question engine and its boundaries	Encoding, geometry, topology, events, identity, persistence, query architecture, GPU/hardware evidence and audit rules.
M198-M257 / KC 2.0	Shape, route and motion laboratory	Parametric patterns, implicit fields, topology descriptors, SE(2)/SE(3), frames, timing, bounded dynamics and hybrid events.
M258-M329 / Two Hands 3.0	World-building and physical interaction	Assets, scenes, geometry compilation, spatial queries, rendering, two-hand transforms, proposals, replay and export.
KC 3.6 definition delta	The rules become objects	Typed definitions, content seals, dependency DAGs, hinges, operation order, definition queries and literal traces.
SCLP 3.6.2 profile	Spiral worlds, route flips and finite addresses	Finite cone/sphere relations, log-polar calculus, jitter guards, phase/winding, reflective wrapping, constraints, grammar and two 64-bit layouts.
SARA 3.6.3 profile	Boundary and permission laboratory (older-teen)	Public/secret typing, checksum and authorization guards, public-only certificates, explicit forbidden scopes and evidence discipline.
M330-M389 / KC Elizabeth 3.9	The playable vector workshop	Vector art, collision, input, animation, tilemaps, procedural audio, deterministic game world, projects, saves and HTML5 export.

STORY -> TOY -> GAME -> FORMAL RECORD

One Idea at Four Learning Depths

Invariant-first translation

The same condition, identity and outcome remain true in every layer. Only the representation changes.

Vocabulary arrives late

Introduce a formal term after the learner can predict and explain the physical behavior.

Return paths stay open

A formal learner can still use the toy to inspect order, ambiguity or a hidden assumption.

Educational design inference. The four-layer method is proposed by this edition; the source reports supply the formal concepts, not a tested educational sequence.

THE THINGS THAT EXIST AND PERSIST

Concept Dictionary I: Rules, State and Identity

Definition

A rule card that says what something means or does.

A typed record with ID, domain, codomain, parameters, dependencies, phase, invariants, provenance and content hash.

Try: Give two door blocks different rule cards and predict their behavior.

Instance

One actual thing using a rule.

A stable node or entity referencing a definition. Many instances may share one definition.

Try: Use one “rolls downhill” card on three different ball tokens.

State

What is true right now.

The current typed values: position, phase, sheet, mode, velocity, ownership and other selected fields.

Try: Flip a bridge tile from closed to open without changing its identity token.

Identity

Who or what this still is, even after it moves or changes.

Stable node ID, generative address, invariants and lineage; coordinates or appearance alone are insufficient.

Try: Move two identical tokens and use lineage cards to tell them apart.

Content seal

A fingerprint that tells us whether the rule changed.

SHA-256 over a canonical definition record, used for integrity and cache identity, not ownership.

Try: Change one word on a rule card and replace its seal sticker.

Evidence boundary

What the game or experiment really proved - and what it did not.

A disposition, scope, uncertainty and non-claim attached to a mechanism or result.

Try: Sort statements into observed, inferred and not established.

Definitions and content addresses: S4 pp. 3-7. Identity split and stable scene nodes: S1 p. 5 and S3 pp. 5-7.

THE SIX-QUESTION ENGINE IN ACTION

Concept Dictionary II: Locality, Matching and Change

Support

The place or group where a rule is allowed to look.

A local admission predicate that prunes irrelevant candidates before relation solving.

Try: A magnet-rule card only applies inside the green hoop.

Compatibility

Two things may touch but still not be allowed to interact.

A predicate over sheet, phase, orientation, mode, policy, ownership or application tags.

Try: Connect only tokens with matching side symbols.

Guard

The “only when...” part of a rule.

A relation crossing or declared condition with accepted status, confidence and error margin.

Try: Open the gate only after the counter reaches three.

Event

The first valid moment when a change is allowed.

The earliest certified guard occurrence after support and compatibility, with degeneracy/conflict status.

Try: Race two timers and record the first valid card.

Transition

The small set of things that change because the event happened.

An atomic bounded patch to scene, route, topology, dynamics or ownership.

Try: Replace one closed tile with one open tile - no other pieces move.

Uncertainty and margin

How sure are we that the token really crossed the line?

Error interval and confidence must remain inside the declared event margin and preserve ordering.

Try: Use a fuzzy boundary strip and decide when the result is “not sure yet.”

Support/compatibility/event sequence: S1 pp. 2-4 and S2 pp. 4-5, 11-12. Interval guards: S5 pp. 5, 9-13.

WHY SPACE, SEQUENCE AND CONNECTION ARE DIFFERENT

Concept Dictionary III: Order, Hinges and Topology

Operation order

Doing A then B may differ from doing B then A.

Phase and composition order are serialized; transforms and hinges need not commute.

Try: Rotate a card then slide it; reset and reverse the order.

Hinge

A place that connects, turns or switches between two maps.

A typed pivot/connector locus with state, selected map, guard and invariants.

Try: Use a zero-value connector piece that changes word or graph order.

Dependency

A rule needs another rule before it can work.

An explicit directed edge in an acyclic definition graph.

Try: Build a chain: power -> sensor -> door, then remove one link.

Cycle

A waits for B, and B waits for A.

An undeclared dependency loop rejected before evaluation unless a bounded cycle policy exists.

Try: Repair a two-card deadlock using an initial-state card.

Topology / gluing

A rule about which edges, routes or sides count as connected.

Explicit port identification with coordinate, sheet and orientation transport.

Try: Tape a paper track with a half twist and move an UP arrow around it.

Winding and route class

Two pieces can arrive at the same place by different journeys.

A covering lift, winding count, homotopy/route descriptor or branch lineage distinguishes re-entry.

Try: Record clockwise and counter-clockwise laps with different route tokens.

Hinges and explicit order: S4 pp. 5-10, 12-14. Route identity and topology maps: S2 pp. 8-9 and S5 pp. 5-10.

THE ADVANCED CONCEPTS WITHOUT THE HYPE

Concept Dictionary IV: Patterns, Packing, Replay and Boundaries

Pattern and field

A repeatable recipe for a path, shape or inside/outside test.

A typed parametric primitive or implicit relation with domain, derivatives and error contract.

Try: Grow a rose or spiral from parameter cards, then test which tokens are inside.

Motion and frame

Where is it, which way is it pointing and how is that changing?

state_at(t), SE(2)/SE(3), moving frames, curvature, bounded timing and dynamic mode.

Try: Move an arrow along a curved track while keeping its local “up” marker consistent.

Seed and bounded grammar

A small starting code chooses a repeatable finite growth recipe.

Seeded branch bits and a budgeted production system with depth, symbol and stack limits.

Try: Use the same bit tiles twice and compare the two gardens.

Key and compression

A compact address tells us where to look, but not everything that exists there.

Finite quantized key layouts, prefix refinement and storage metrics with reconstruction/error contracts.

Try: Use a coordinate code to find a drawer, then list the information the code does not contain.

Proposal, commit and replay

Try a change in a pretend world, check it, then make it real and remember it.

Candidate patch -> verification/conflict policy -> deterministic atomic commit -> pre/post hashes and replay.

Try: Preview a card sequence on a spare board before updating the main board.

Permission and secret boundary

Being able to see something does not mean you may use or change it.

Typed public/secret partition, authorization guard, zero secret egress and explicit forbidden scopes.

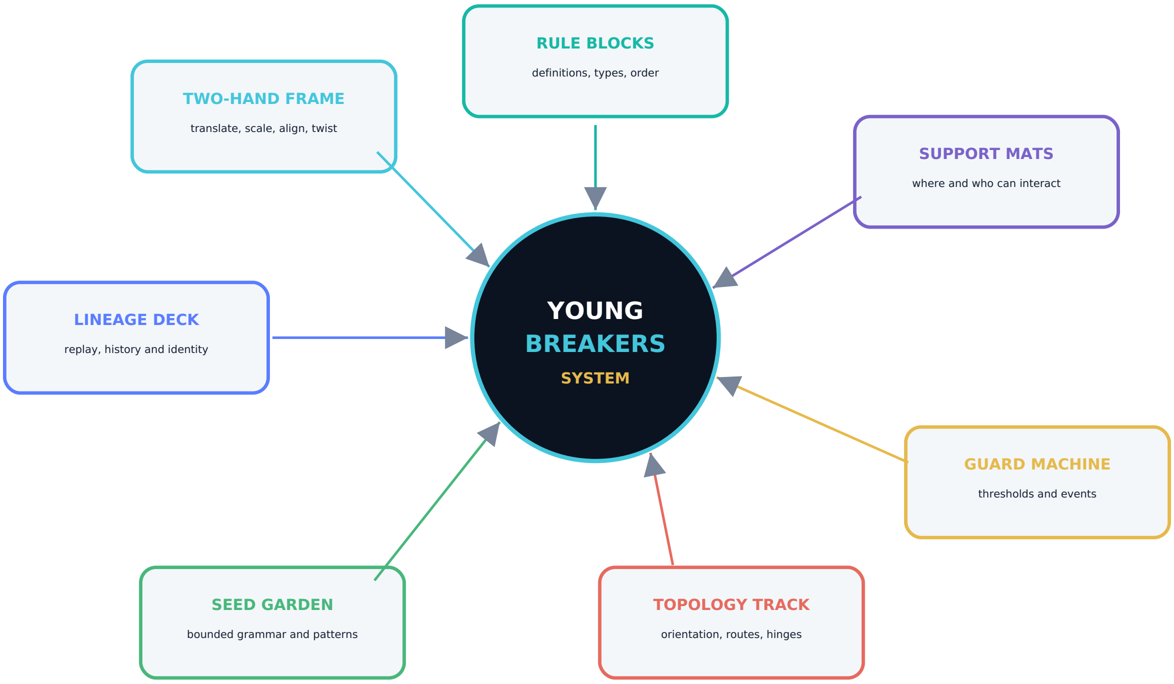
Try: Sort public instruction cards, private answer cards and actions that require permission.

Older-teen SARA boundary. The permission lesson must remain abstract or use harmless classroom fixtures. It must not include real wallet secrets, key search, signing, transactions, balance targeting or ownership claims.

Patterns/kinematics: S2 pp. 5-12. Packing/grammar: S5 pp. 7-14. Proposal/replay: S3 pp. 11-16. Authorization and secret partition: S7 pp. 1-10, 13-15.

A FAMILY OF COMPATIBLE MANIPULATIVES

Physical Learning Ecosystem



PHYSICAL FIRST, DIGITAL LATER

Shared visual grammar

Every product reuses the same colors, icons, tactile marks and child questions so knowledge transfers between kits.

Non-electronic first

The starter pathway must work with paper, board, wood or molded pieces. Digital identification remains optional.

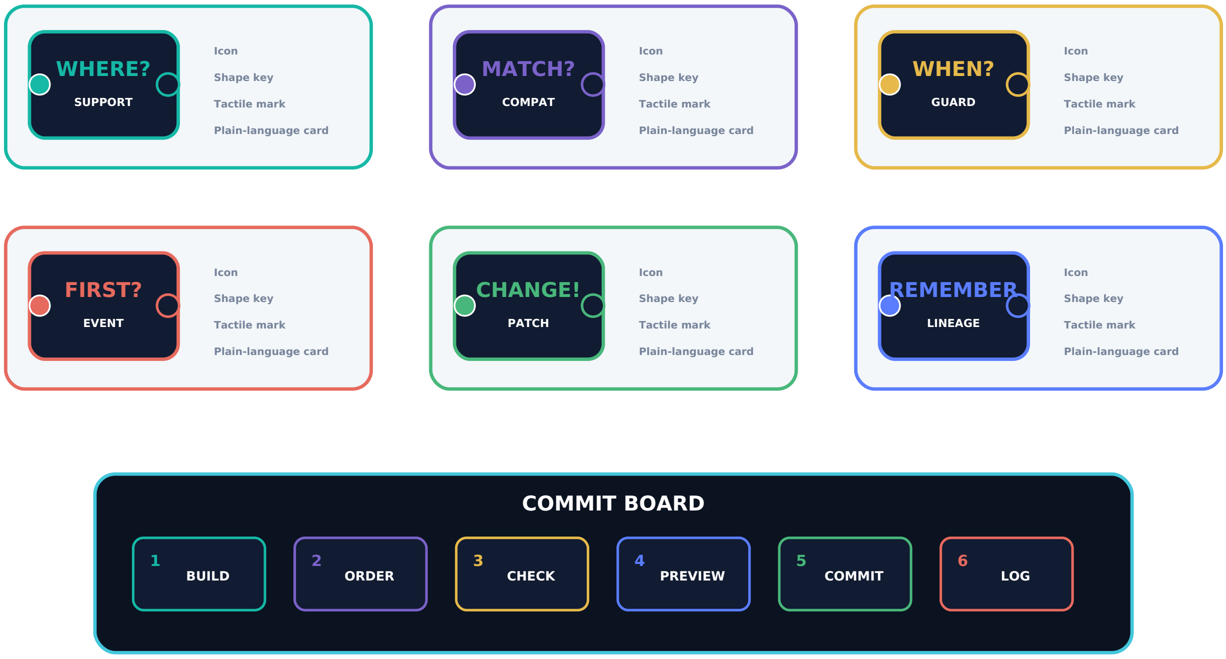
Verifier as a role

Before software performs verification, a child or adult can play the verifier and explain each gate aloud.

Product inference. The ecosystem is a scaffold for prototyping. Component dimensions, materials, certifications, costs and market positioning remain intentionally unfilled.

THE SUBSTRATE IN THE HAND

Toy Scaffold 1: Rule Blocks and Commit Board



PLACEHOLDER: magnet-free keyed sockets, thick cards, optional wooden or molded blocks, no electronics required.

LEARNING GOAL

Definitions, types, dependencies, operation order, preview and commit.

EARLY PROTOTYPE

Print-and-cut cards with large keyed tabs and a six-zone paper board.

LATER FORM

Chunky blocks with tactile class marks; material and socket system TBD.

SUCCESS EVIDENCE

Learner predicts an order effect and explains one rejection.

NON-GOAL

Do not simulate a general programming language or expose raw JSON.

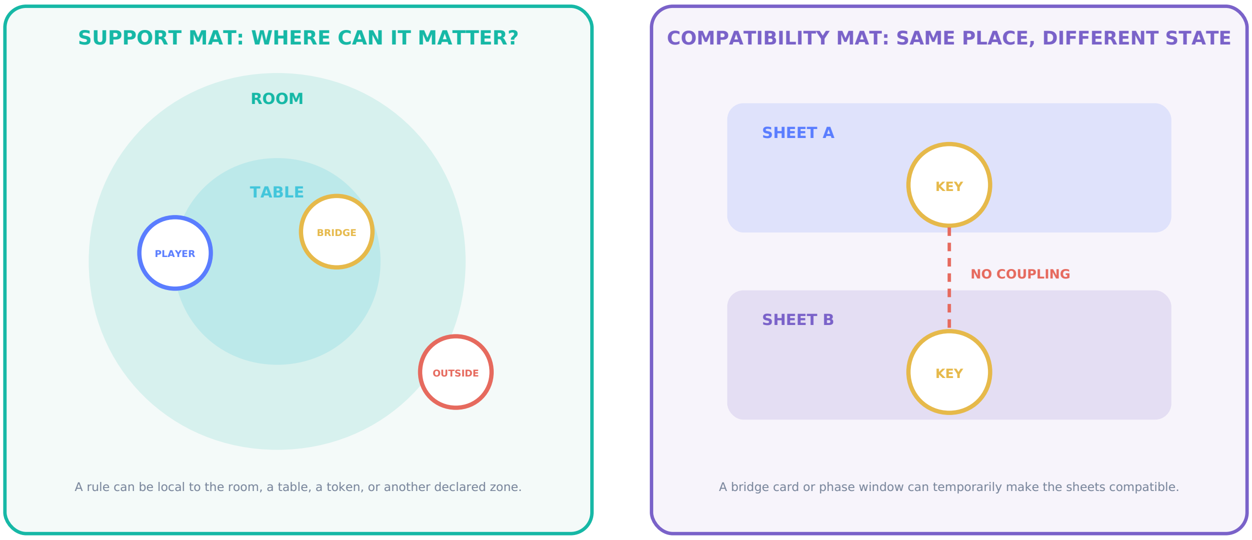
OPEN QUESTIONS

Best socket shapes, block count, storage tray, age grading and durability.

Definitions, dependency closure and operation order: S4 pp. 3-10. Proposal/commit: S3 pp. 11-13.

LOCALITY BEFORE INTERACTION

Toy Scaffold 2: Support and Compatibility Mats



PLACEHOLDER MATERIALS: foldable cloth or board mats, chunky tokens, raised outlines, high-contrast icons.

Starter activity: Where can it matter?

1. Place a rule card beside a marked zone.
2. Move several tokens around the mat.
3. Apply the rule only to tokens inside the declared support.
4. Ask what work was avoided by ignoring the rest.

Progression: Same place, different state

1. Stack two transparent sheets or use two adjacent layers.
2. Place identical tokens at matching coordinates.
3. Show that interaction is blocked until a compatibility bridge is played.
4. Add orientation, phase or ownership tags later.

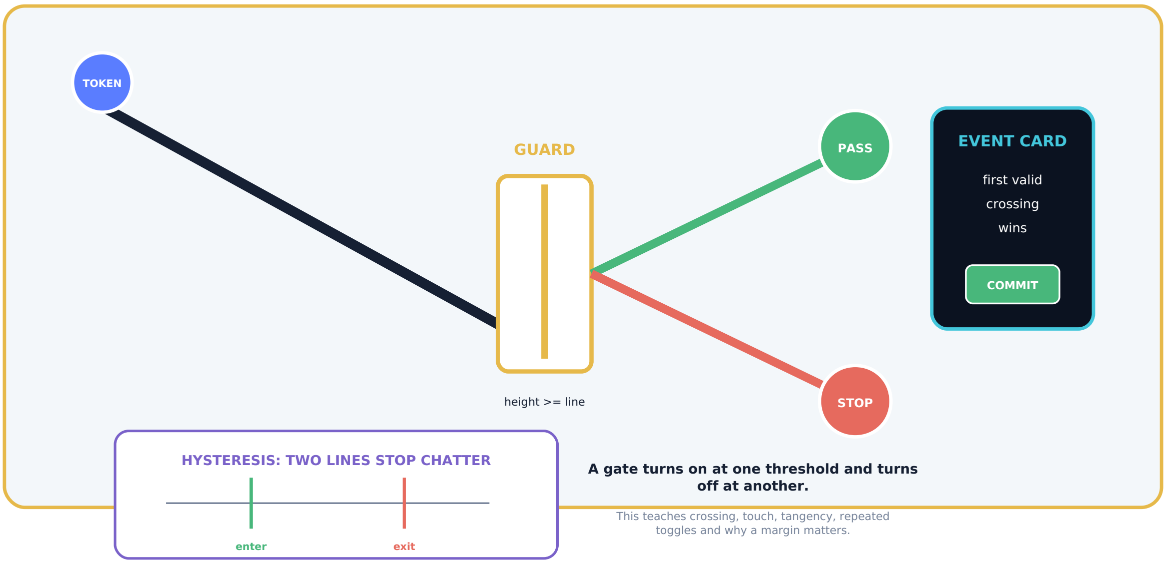
Accessibility rule. Sheet, support and compatibility must use shape, texture and labels as well as color; transparent overlays need an opaque alternative.

Same-coordinate/different-sheet and compatibility boundaries: S1 pp. 4, 18-19; S2 pp. 4-5, 8; S5 pp. 5-9.

MAKE “WHEN” VISIBLE

Toy Scaffold 3: Guard Gate and Event Machine

GUARD GATE AND EVENT MACHINE



This teaches crossing, touch, tangency, repeated toggles and why a margin matters.

PLACEHOLDER: use large balls/tokens only; no small-part version for early-years kits.

Basic lesson

A rule waits until a clearly marked threshold is crossed. Before the line: no event. After the line: one event.

Intermediate lesson

Compare crossing, touching and sliding along the boundary. Introduce “not sure yet” when a token lies inside a fuzzy margin.

Advanced lesson

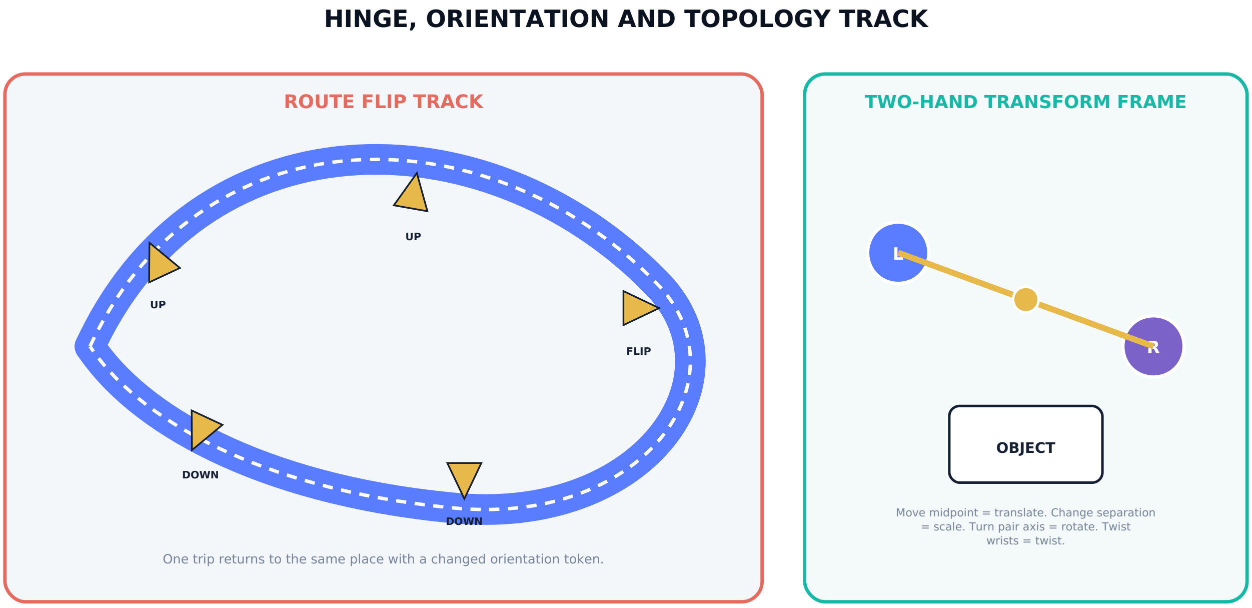
Use separate enter/exit lines to stop chatter, or race two events and apply an explicit priority/ conflict rule.

Safety placeholder. Early-years versions should avoid small balls and spring-loaded parts. Final component sizes, pinch hazards, materials and certification requirements must be determined by qualified toy-safety work.

Guard bands, hysteresis and event degeneracy: S1 pp. 18-19; S2 pp. 11-12; S5 pp. 5, 13-14.

CONNECTION, ORIENTATION AND EMBODIED TRANSFORMATION

Toy Scaffold 4: Hinge, Topology Track and Two-Hand Frame



PLACEHOLDER: cardboard/wood frame with sliding handles and orientation arrows; no tracking hardware required.

Route Flip Track

A token carries a large orientation arrow. Traversing a twisted connection changes that arrow according to an explicit rule. This makes topology a routing transformation rather than a mysterious shape trick.

Two-Hand Transform Frame

Two handles control midpoint, separation, pair-axis direction and twist. The learner feels translation, scale, alignment and rotation as different quantities before seeing them as a transform record.

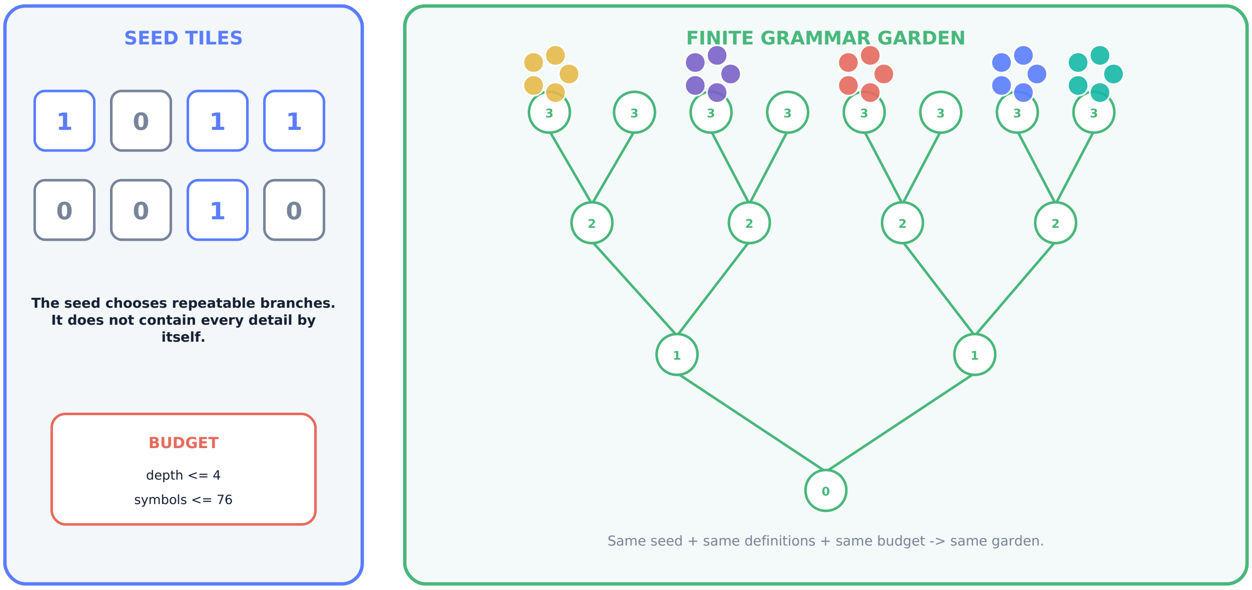
Physical boundary. The track is a mathematical routing manipulative. It does not claim physical Klein-bottle self-assembly, energy generation or impossible material behavior.

Explicit gluing and orientation: S2 pp. 8-9; S4 pp. 5-6; S5 pp. 5-9. Bimanual transform: S3 pp. 11-12.

SMALL BEGINNINGS, FINITE GROWTH

Toy Scaffold 5: Seed Garden and Pattern Builder

SEED GARDEN AND BOUNDED PATTERN BUILDER



PLACEHOLDER: branch tiles, turn cards, reusable stems and a budget ruler; optional camera recognition later.

Repeatability

Two groups start with the same seed, production cards and budget. They should grow the same structure if they apply the rules in the same order.

Bounded growth

A physical depth ruler and token budget stop the grammar. Overflow is a visible result, not a hidden crash or promise of infinity.

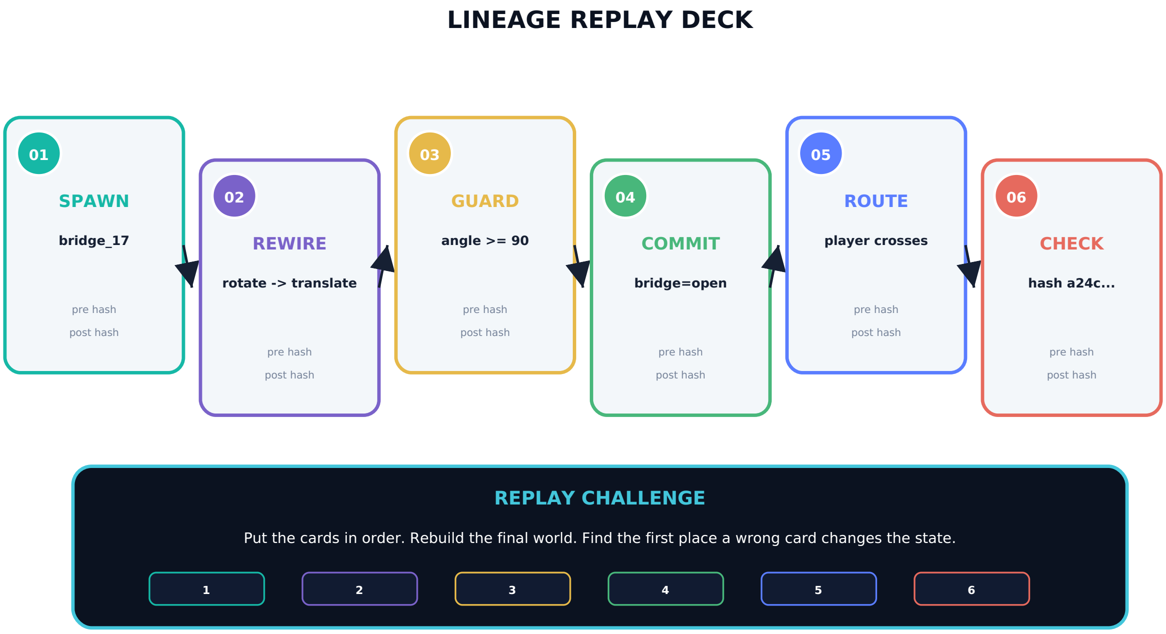
Compression honesty

The seed selects branches, but the kit still needs rule definitions, budgets and any external changes. A seed is not the whole world.

Pattern families and bounded grammar: S2 pp. 5-7, 12-15; S5 pp. 7-14. Seed/reconstruction boundary: S1 pp. 12-13 and S6 pp. 9, 12.

REMEMBER ENOUGH TO REBUILD AND COMPARE

Toy Scaffold 6: Lineage Replay Deck



PLACEHOLDER: thick sequence cards, dry-erase hash/identity fields, optional QR codes only in later prototypes.

Replay as explanation

Learners rebuild a final board from ordered event cards. A mismatch reveals the first point where state diverged.

Identity beyond appearance

Two identical-looking objects can be distinguished by their stable IDs and lineage histories. A changed asset or coordinate does not erase the instance.

Card field	Child-facing use	Formal counterpart
Sequence number	Where this card belongs	Monotonic event order
Who changed	Object identity token	Stable node/entity ID
Before / after	Two simple state pictures	Pre/post state or hashes
Why	Reason icon and sentence	Guard status, proposal type and reason code
Route/history	Journey mark	Lineage, winding or branch label

Lineage and identity: S1 p. 5; S2 pp. 8-9, 13; S3 pp. 6, 13-16; S6 p. 9.

TABLETOP PROOF BEFORE SCREEN COMPLEXITY

Physical Games Before the Digital Game

PHYSICAL GAME FAMILY

RULE RELAY
Teams pass typed blocks down a phase rail. Wrong order creates a visible failure.

order + dependencies

SAME PLACE, DIFFERENT SHEET
Move tokens on stacked maps. Co-located pieces interact only through compatibility windows.

topology + compatibility

EVENT DETECTIVE
Inspect a sequence of threshold clues and name the first valid event.

guards + event order

SEED GARDEN
Grow a bounded branching pattern from shared seed tiles and compare repeatability.

grammar + seeds

REPLAY RESCUE
Repair a scrambled lineage deck so the final world hash matches the goal.

lineage + replay

COMMIT OR REJECT
Players propose edits; a rotating verifier role checks support, compatibility, guard and invariants.

whole authority chain

All games are scaffolds: final rules, age grading, timings, component counts and safety tests remain to be prototyped.

Cooperative default
Players can share a goal while rotating the verifier role. This separates proposing a bold idea from checking whether it is valid.

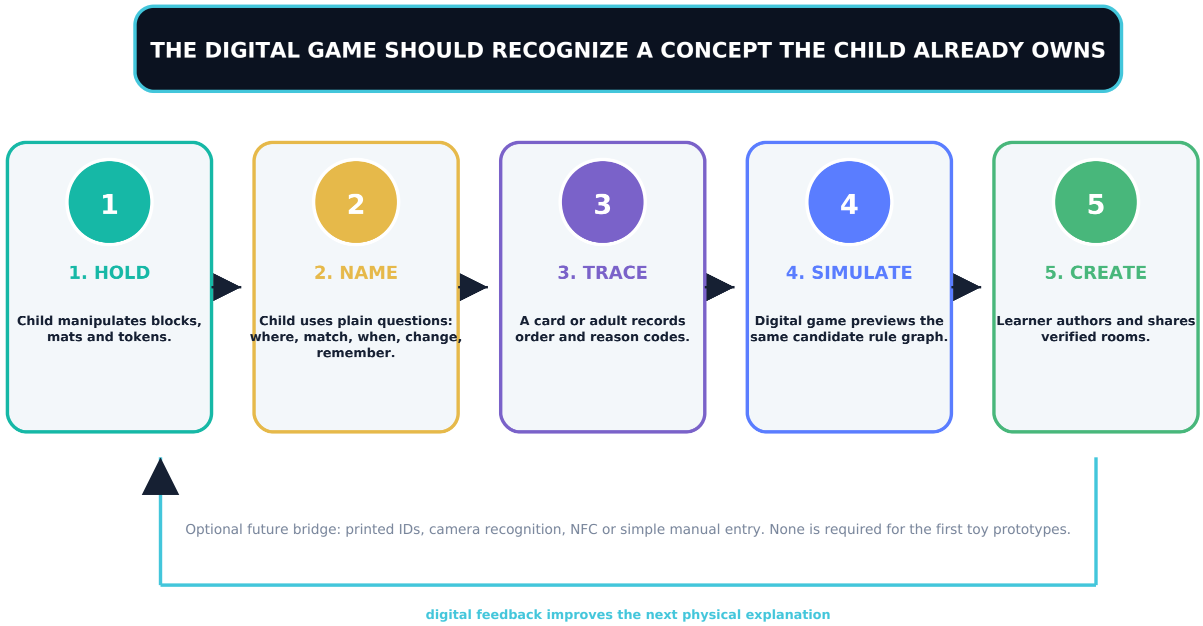
Competitive variants
Race, deduction and optimization modes may be added only when they do not reward hiding definitions or exploiting unclear rules.

Proof of learning
A successful turn includes a prediction and explanation, not only a winning move.

Game-design scaffold. Turn structures, balance, player counts, durations and component counts require iterative prototyping. The titles and mechanics on this page are not finished products.

THE SCREEN SHOULD REVEAL, NOT REPLACE, UNDERSTANDING

From Physical Understanding to Digital Mastery



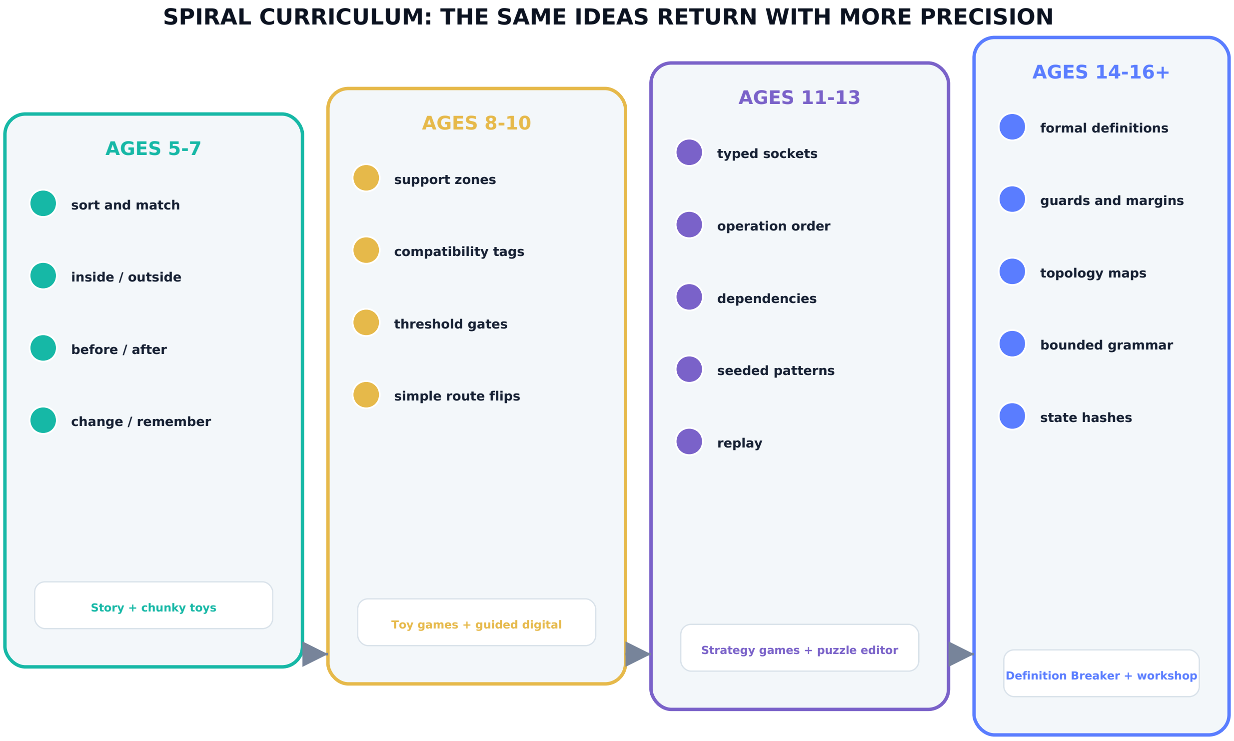
Digital step	Working game concept	Primary learning transfer
1	**Support Scouts** - sort moving objects into local rule zones.	Physical mats become animated supports and reason-coded rejection.
2	**Socket Match** - connect only compatible shapes, modes and phases.	Keyed toy connectors become typed digital sockets.
3	**Guardians of the Gate** - predict threshold crossings and chatter.	The guard machine becomes a visible timeline and interval preview.
4	**Route Flip** - navigate orientation-changing sheets.	The physical track becomes topological rooms and route lineage.
5	**Order Lab** - reorder transforms and dependencies.	The Commit Board becomes the digital phase rail and trace inspector.
6	**Definition Breaker Academy** - author verified mini-rooms.	All manipulatives converge on the full rule-object editor.
7	**Definition Breaker campaign**	The learner uses the same concepts inside the complete puzzle-adventure.

Privacy/offline boundary. The first digital companions should work offline and should not require accounts, cameras, NFC or cloud services. Recognition technologies are optional later experiments with explicit consent and data handling.

The practical digital target is the self-contained 3.9 Canvas/Web Audio runtime with named input actions, deterministic simulation and local persistence (S6 pp. 3-12, 15).

RETURN TO THE SAME CONCEPTS WITH RICHER TOOLS

Spiral Curriculum and Age-Band Progression



Age bands are starting points, not ability labels. Learners can move sideways, revisit and mix layers.

Return, do not rush

The same distinction should reappear in new media and harder contexts rather than being "covered" once and abandoned.

Move sideways

A learner may use formal language with physical pieces or revisit a story layer to resolve a difficult abstraction.

Show transfer

Progress is demonstrated when the learner recognizes and explains the concept in a new toy, tabletop or digital situation.

Curriculum status. The age bands and spiral are instructional design proposals. Sequence, duration, age suitability and outcomes require playtesting and qualified educational review.

PUT THE SPIRAL INTO PRACTICE

Sample Learning Units and Session Flow

Unit A - Where and who?

Tools: support mat, compatibility tokens.
Goal: explain why two nearby objects do or do not interact.
Evidence: learner predicts three cases and corrects one misconception.

Unit B - When does change happen?

Tools: guard machine, threshold cards.
Goal: distinguish before, crossing, touch and after.
Evidence: learner uses "not sure yet" inside a margin.

Unit C - Does order matter?

Tools: rule blocks, Commit Board.
Goal: compare A->B with B->A.
Evidence: learner locates the first divergence.

Unit D - Can we rebuild it?

Tools: lineage deck, digital replay.
Goal: reconstruct state from an ordered log.
Evidence: learner identifies the first wrong card/hash.

A repeatable 30-60 minute session scaffold

1. Story prompt
2. Predict
3. Manipulate
4. Verify aloud
5. Replay
6. Transfer

Evidence to capture

Prediction, first explanation, reason for a rejection, revised explanation, and whether the learner transferred the idea to a new representation.

Adult role

Ask questions, keep candidate and committed states separate, and supply formal vocabulary only after the learner can describe the behavior.

Session boundary. The durations and lesson sequence are placeholders. They are not validated classroom plans or therapeutic/diagnostic activities.

FACILITATE EXPLANATION, NOT MEMORIZATION

Educator and Parent Guide

Ask prediction questions

- Where is the rule allowed to act?
- Which tokens are compatible, and what tag proves it?
- What exact condition makes the change legal?
- What will change, and what must stay the same?
- How will we remember or replay what happened?
- What did our experiment not prove?

Normalize rejection

A rejected edit is not failure; it is a reason-coded observation. Ask the learner to name the failed gate and propose one change to the definition, support, compatibility, guard or order.

Keep authority visible

Use a separate preview board or ghost state. Never quietly change the main board while discussing a candidate.

Common misconceptions

Misconception	Prompt
“They touch, so they interact.”	Which compatibility tag is missing?
“The key code contains the whole world.”	Which definitions and external events are still needed?
“The picture proves the rule.”	Is this rendering authoritative or a projection?
“One bit is all the state.”	What continuous values, uncertainty and identity are separate?
“A strange shape creates energy.”	Where is the physical model, input and measured loss?
“A checksum proves ownership.”	What does it actually validate, and what permission is absent?

Observation rubric

1. Recognizes

Names or points to the relevant object, zone or rule class.

2. Predicts

States what will happen before the move or commit.

3. Explains

Uses the child question or formal term to justify the result.

4. Transfers

Applies the same idea in a new toy, tabletop or digital context.

Assessment boundary. This rubric is formative, not a standardized test or diagnostic tool.

NO CONCEPT SHOULD DEPEND ON ONE SENSE OR ONE INPUT

Inclusive Play: Multiple Ways to Understand

Minimum access rule: no core concept may depend on one sense, one motor action, fast timing, fluent speech or advanced reading. A child must be able to understand and answer through at least two different routes.

Visual

- Color is always paired with icon, word and border pattern.
- High-contrast and low-glare print variants.
- Large state-change arrows and uncluttered layouts.
- Opaque alternatives to transparent sheet overlays.

Tactile and motor

- Raised class marks and distinct edge shapes.
- Chunky pieces, stable mats and low-force connectors.
- One-handed alternatives to two-hand activities.
- Turn-based modes without speed pressure.

Language and cognition

- Plain question first, formal name second.
- One new distinction per early activity.
- Picture, spoken and written instructions.
- First-divergence hints instead of dense traces.

Same concept, different modality

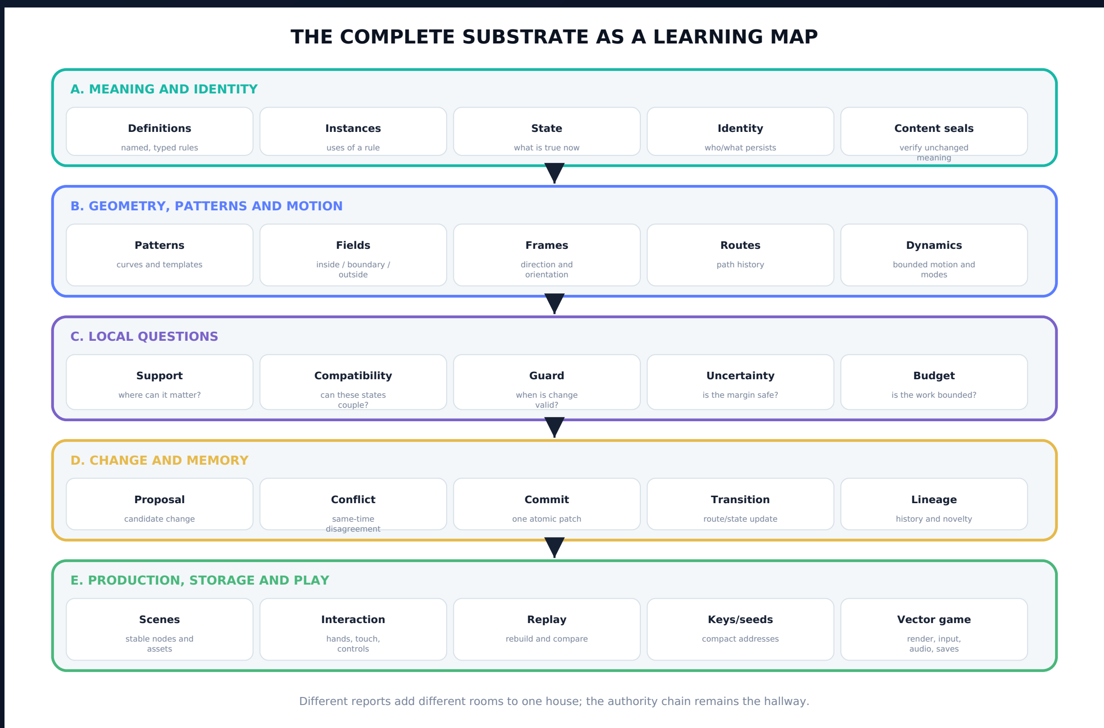
Concept	See it	Touch it	Hear it	Say it
Support	Colored/outlined zone	Raised boundary mat	Admission click	“Where can it matter?”
Compatibility	Matching symbols	Keyed connectors	Two tones join	“Are they allowed to match?”
Guard	Threshold line	Gate detent	Rising cue / crossing chime	“When is change allowed?”
Commit	Preview becomes solid	Move candidate tile to main board	Commit chord	“Make one checked change.”
Lineage	Numbered trail	Ordered cards	Replay motif	“What do we remember?”

Accessibility design status. These are required directions for prototyping, not evidence that a final product has been tested with every learner or access need.

PART III - ACCESSIBLE BY DEFAULT

ACCESS IS PART OF THE RULE SYSTEM

Digital, physical, language, sensory and co-design requirements for children with different bodies, senses, communication styles and learning routes.



Design-derived accessibility specification. It must be tested with children and families; this document does not certify universal access.

DEFAULTS BEFORE DIFFICULTY

Digital Accessibility Must Be Available at First Launch

Architecture rule: access options are loaded before the first puzzle, stored separately from difficulty and preserved in saves and replays without changing puzzle authority.

DEFAULT

Readable presentation

- Text scale and zoom before play.
- High-contrast, low-glow and thick-line modes.
- Icon + word + border pattern for every rule class.
- No tiny particles or color-only state.

DEFAULT

Safe pace

- Full pause at every Commit Bench.
- No timer in Guided mode.
- Adjustable animation, trace and replay speed.
- Unlimited safe previews and checkpoint undo.

DEFAULT

Choice of input

- Keyboard, gamepad, pointer and touch mappings.
- One-hand layouts and toggle alternatives.
- Snap strength, dwell time and repeat delay.
- Partner-assisted and switch-style step selection.

DEFAULT

Choice of explanation

- Picture and plain-language layer first.
- Optional narration and captions.
- First failed question highlighted.
- Formal trace and full record remain optional.

Source-supported platform basis: the 3.9 project model already uses named actions across keyboard, gamepad, pointer and touch, deterministic input frames, Canvas presentation, procedural audio, persistence and offline HTML5. The access behavior above is an application requirement, not a claimed feature of the supplied runtime.

ONE IDEA, MANY CONTROL PATHS

Motor and Input Access

Need	Required design response	Acceptance check
One hand	Every core action can be completed without simultaneous two-hand input. Two-hand activities have sequential or partner-assisted equivalents.	Complete the child route and first four puzzle rooms using one side of a controller or one touch region.
Low force / limited reach	Large hit areas, adjustable snap distance, no drag endurance requirement, no rapid repeated presses.	All sockets activate with the largest target preset and no hold longer than the chosen threshold.
Hold is difficult	Every hold action has toggle, latch or confirm alternatives. Pinch/hold does not gate understanding.	Run the full loop using toggle mode only.
Precise pointing is difficult	Directional navigation, ordered socket stepping, undo and focus highlight.	Complete a room without free pointer movement.
Single-switch route	Step through relevant choices, pause on selection, confirm or cancel with configurable scan timing.	A scripted scan path can reach every required command without entering expert-only controls.
Fatigue or variable control	No progress loss for pausing; adjustable repeat delay; optional automatic cable routing; short interaction chunks.	Pause at any step, resume from the same candidate state and finish without timing penalty.

Do not infer ability from speed. Completion time is never used as the only evidence of understanding.

Native OpenXR or hand tracking may be added later, but it must never become the only route. The first release should remain fully operable through conventional named actions.

MEANING SURVIVES PRESENTATION CHANGES

Vision, Reading and Color Access

MEANING

Never color alone

Every class and state uses at least three of: word, icon, border shape, pattern, position, sound or tactile code. Candidate and committed states remain distinguishable in monochrome.

READING

Plain first, formal second

“Are they allowed to match?” appears before “compatibility.” Sentences are short, instructions are chunked and jargon unlocks in context.

LOW VISION

Scale without losing the puzzle

Text, trace thickness, pointer size and object outlines scale independently. The camera can zoom while the Rule Lens presents one focused object.

ASSISTIVE TECH

Semantic labels

Rule blocks, sockets, phases, reason codes and buttons expose stable names, current values, state and order. Decorative particles are hidden from reading order.

TEST

Monochrome proof

Print the first activity in grayscale. A new participant can still distinguish every rule class and current/preview/committed state.

TEST

Large-text proof

At the largest supported text preset, no essential control is clipped, covered or moved off-screen; a simplified single-column inspector is available.

PDF note: this edition uses searchable text, document language, bookmarks, image descriptions and tagged structure. Independent assistive-technology and PDF/UA testing is still required.

NO AUDIO-ONLY CLUE AND NO SPEECH-ONLY ANSWER

Hearing, Speech and Communication Access

Game output

- All dialogue, alerts and rule cues have captions or visible equivalents.
- Speaker, direction and cue type are identifiable without sound.
- Music, effects, voice and confirmation sounds have separate controls.
- A waveform or icon may show rhythm puzzles; audio is optional.
- Optional narration reads plain-language instructions and reason codes.

Communication board strip

WHERE - MATCH - WHEN - YES - CHANGE - REMEMBER

Also include: **again**, **pause**, **help**, **not that one**, **my turn**, **stop**.

Child response

- Pointing, eye-gaze selection, card choice, switch input, gesture, drawing and partner-assisted movement are valid responses.
- Speech recognition is never required for core progress.
- Open-ended explanation may be replaced by ordered picture cards or before/after choices.
- A communication partner confirms the child's selection without silently changing it.
- Co-op roles rotate so non-speaking players are not confined to passive observation.

Authority rule: the child's chosen response commits the action. The helper is an input adapter, not the decision-maker.

Language versions need a versioned glossary and child review. Translation must preserve the six-question meaning rather than copy formal terminology blindly.

PREDICTABLE, REVERSIBLE AND RESPECTFUL

Sensory and Emotional Safety

Failure language: use “not yet,” “first check,” or “this promise would break.” Never shame the child, erase earned progress or use a startling effect as the explanation.

Reduced motion

- Replace camera shake, zoom bursts and particles with static outlines or arrows.
- Slow or step through transformations.
- Preview motion intensity before play.

Reduced sound

- Quiet preset at first launch.
- No surprise loudness.
- Separate warning, success, music and ambience controls.

Predictability

- Show session steps and expected duration.
- Warn before new mechanics, boss changes or sensory effects.
- Keep buttons and rule classes in stable places.

Safe pause and exit

- Pause immediately from any state.
- Save the candidate graph without forcing a commit.
- Exit and return without punishment or lost access choices.

Processing time

- No countdown by default.
- Instructions remain visible.
- One new distinction per early activity.

Recovery

- Undo by checkpoint and replay.
- Show what stayed correct.
- Offer the smallest next clue, not the full solution.

These are design safeguards, not clinical treatment claims. Individual children and families determine which options help.

THE SAME CONCEPT, DIFFERENT PHYSICAL FORMS

Accessible Physical Kit Variants

Variant	Placeholder design	Concepts preserved	Safety / validation work
Large-piece kit	Chunky blocks, wide sockets, stable trays, large print and thick borders.	Type, order, support, compatibility and commit.	Dimensions, mass, reach, drop and pinch testing.
Tactile kit	Raised class marks, distinct edges, textured zones and ordered notches.	Rule class, socket match, phase order and lineage.	Tactile legibility with blind/low-vision users; cleaning and wear.
Opaque sheet kit	Side-by-side boards or hinged layers instead of transparent overlays.	Same coordinate, different sheet and compatibility bridge.	Alignment, contrast and accidental-move testing.
Low-force / one-hand kit	Velcro-style or gravity-held pieces, anti-slip mat, card rack and turntable.	All core questions without two-hand grip or strong connectors.	Force, stability, reach and fatigue evaluation.
No-magnet kit	Keyed cardboard, foam or mechanical slots; no embedded magnets.	Typed connections and incompatibility rejection.	Material, small-part and durability review.
Quiet / low-sensory kit	Soft-contact pieces, matte surfaces, muted optional feedback and predictable storage.	Guard, commit and replay without startling cues.	Noise, glare, odor, texture and cleaning evaluation.

Prototype rule: never shrink an accessible variant into a token “special” kit. The same puzzle goals, meaningful choices and progression must remain available.

All physical concepts remain placeholders. Manufacturing, age grading, small-part, magnet, material, chemical, pinch, cleaning and regulatory decisions require qualified review.

BUILD WITH CHILDREN, NOT ONLY FOR THEM

Co-Design, Playtest and Accessibility Acceptance

1. LISTEN

Use child assent, caregiver context and preferred communication. Ask what helps before showing a solution.

2. OBSERVE

Record where meaning, control, sensory load or language blocks participation. Do not diagnose.

3. CHANGE

Alter the rule, interface, physical component or facilitation - not the child.

4. VERIFY

Retest with the same access route and a new participant. Preserve puzzle meaning and child authority.

Participant coverage for prototypes

Include	What to learn	Stop or redesign when
Children with varied vision, hearing, motor control, communication, reading and sensory preferences	Whether the same concept can be perceived, chosen and explained through different routes.	A core idea disappears when one sense or input route is removed.
Children across the proposed age bands and prior game experience	Whether language, choice count and abstraction are understandable without oversimplifying.	Success depends on hidden prior knowledge, reading level or adult translation.
Caregivers, educators and accessibility specialists	Setup burden, partner behavior, fatigue, safety, communication and transfer between physical and digital play.	The adult must take over the child’s decisions or constantly repair the interface.

Minimum release gates

GATE

Perception

All required states are distinguishable without color and without sound.

GATE

Operation

The first four rooms are completable one-handed and without precise free dragging.

GATE

Understanding

A child can answer each big question through speech or a non-speech route.

GATE

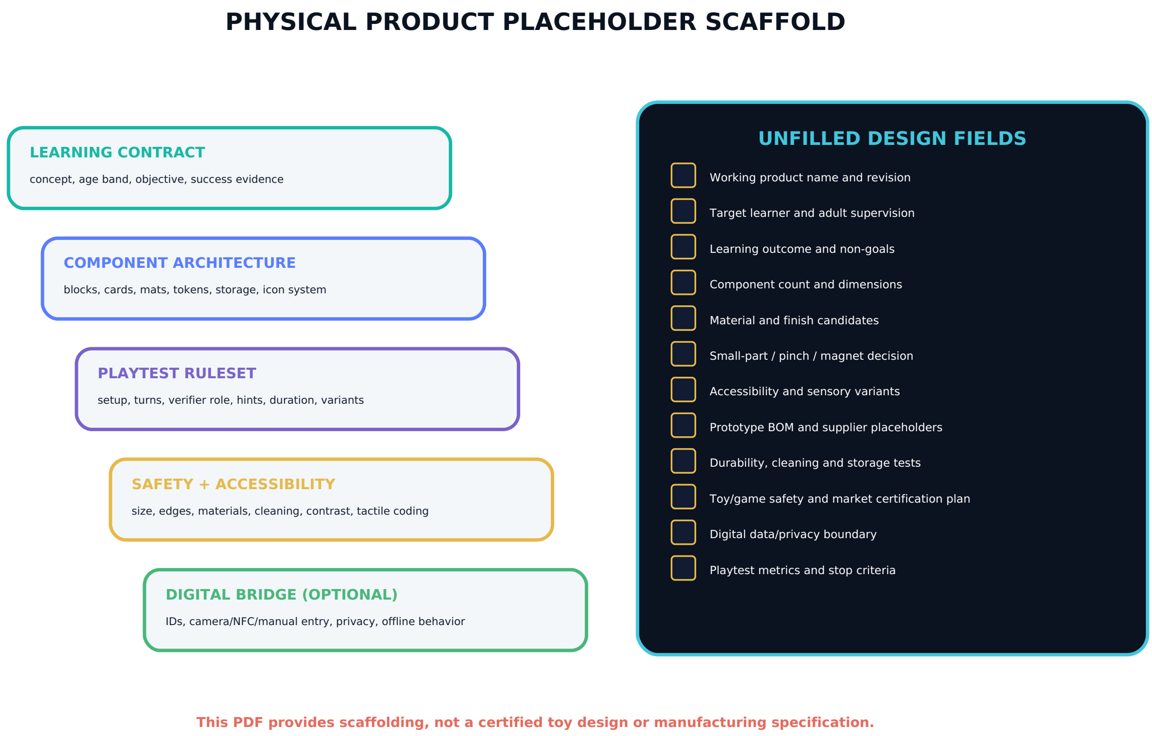
Well-being

Pause, stop and sensory-reduction choices work immediately and preserve progress.

This protocol is formative design scaffolding, not a standardized assessment, clinical study or universal-access certification.

BUILD THE EVIDENCE BEFORE THE PRODUCT

Physical Product and Safety Placeholder Scaffold



Prototype sequence

1. Paper/card proof of learning loop.
2. Large low-cost foam/cardboard manipulatives.
3. Material and connector experiments.
4. Accessibility and durability variants.
5. Qualified safety review and age grading.
6. Only then: manufacturing engineering and certification.

Explicitly not delivered here

- Final dimensions, material specifications or tolerances.
- Small-parts, magnet, pinch or chemical safety assessment.
- Costed BOM, tooling, supplier or packaging specification.
- Regulatory certification or market-compliance claim.
- Production CAD, mold design or electronics design.

Physical-product boundary. Every toy section in this PDF is conceptual scaffolding. A manufacturable product requires qualified mechanical, material, safety, accessibility, educational and regulatory work.

THE NEXT PROOF IS UNDERSTANDING, NOT PRODUCTION VOLUME

Young Breakers Roadmap and Acceptance Gates

Recommended development path

0. Vocabulary test

Test the six child questions with paper examples. Stop if adults must translate every turn.

1. Three physical activities

Rule Blocks, Support Mat and Guard Gate. Capture predictions, explanations and misconceptions.

2. Tabletop proof

Prototype Rule Relay, Same Place/Different Sheet and Replay Rescue with rotating verifier roles.

3. Digital companion

Implement Support Scouts, Socket Match and Order Lab in the 3.9 vector runtime.

4. Academy + campaign

Connect the learning path to the Definition Breaker vertical slice and workshop authoring.

Acceptance gates

Gate	Pass condition	Stop/simplify condition
Concept transfer	Learner recognizes the same rule in toy and digital forms.	The digital layer introduces a new metaphor unrelated to the physical activity.
Explanation	Learner can state where, match, when, change and remember for a solved task.	Success depends on trial-and-error with no causal explanation.
Rejection quality	Invalid moves produce a specific, useful reason.	Feedback is generic, punitive or visually ambiguous.
Boundaries	Uncertainty, non-claims, finite budgets and permission remain visible.	The product rewards hype, hidden state or unsafe overgeneralization.
Physical viability	Qualified reviewers can define safe materials, dimensions and certifications.	The learning goal requires hazardous, fragile or excessively complex components.

ROADMAP DECISION

Do not begin with molded toys or a general-purpose editor.

First prove that the six child questions improve prediction, explanation and transfer using paper prototypes and a narrow digital companion.

FINAL GROUNDING AND EVIDENCE BOUNDARY

Source Register, Accessibility Status and Upgraded Vision

Source register for Part II

ID	Source role
S1	**UGTS-GN 1.1**: canonical query sequence, identity, one-bit and evidence boundaries.
S2	**UGTS-KC 2.0**: patterns, fields, topology, motion, event classes and bounded dynamics.
S3	**KC Two Hands 3.0**: physical manipulation, stable nodes, proposals, commit and replay.
S4	**UGTS-KC 3.6**: definitions inside the substrate, dependency graphs, hinges and operation order.
S5	**UGTS-KC 3.6.2 SCLP**: winding, reflective wraps, interval guards, constraints, grammar and finite keys.
S6	**UGTS-KC 3.9**: practical vector game, input, simulation, feedback, persistence and HTML5.
S7	**UGTS-KC 3.6.3 SARA**: optional older-teen permission, public/secret and authorization boundary.
S8	**Definition Breaker DB-0.1**: original game concept expanded by this edition.

Source-derived: formal concepts, version lineage, implementation boundaries and reported validation within each supplied document.

Design-derived: pedagogy, age bands, stories, toy system, activities, game titles, accessibility directions and product roadmap.

DB-0.3 accessibility delta: easy-read child route, choice-based access passport, multimodal digital requirements, one-handed and switch-compatible input scaffolds, sensory safety, accessible physical-kit variants, co-design protocol and acceptance gates.

UPGRADED VISION

Definition Breaker becomes a lifelong, accessible learning ladder.

A child can first hold a boundary, point to a match, hear a guard, choose a change and replay what happened. Years later, the same learner can inspect typed definitions, topology and uncertainty without abandoning that first intuition.

THE CORE PROMISE

Every impossible-looking change becomes understandable before it becomes authoritative.

Story, toy, tabletop and digital layers all preserve this promise.

Final evidence and accessibility boundary. Formal concepts and version lineage are source-derived. Learning, accessibility and physical-product proposals require participatory research with disabled and non-disabled children, caregivers, educators and specialists, plus qualified safety and regulatory review. The PDF is tagged and searchable, but no independent conformance certification is claimed.